2018 Level I Mock Exam (B) PM

The afternoon session of the 2018 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

Questions	Topic	Minutes
1–18	Ethical and Professional Standards	27
19–33	Quant	22.5
34–45	Econ	18
46–69	Financial Reporting and Analysis	36
70–78	Corporate Finance	13.5
79–86	Portfolio Management	12
87–98	Equity	18
99–110	Fixed Income	18
111–115	Derivatives	7.5
116–120	Alternative Investments	7.5
	Total:	**180**

2018 LEVEL I MOCK EXAM (B) PM

1 Claire Jones, CFA, is an analyst following natural gas companies in the United States. At an industry energy conference, the CFO of Alpine Energy states they are interested in making strategic acquisitions. At a separate event, Alpine's head of exploration commented he is bullish on natural gas production prospects within Northeastern Pennsylvania. Jones is aware that Alpine currently has very little exposure to this region. She also knows another company in her universe, Pure Energy, Inc., is based in Northeastern Pennsylvania and controls significant assets in the area. Pure Energy is highly leveraged, and Jones believes they will need to raise additional capital or partner with another firm to move to the production phase with their assets. Jones attempts to contact Alpine's CEO with an unrelated question and is told he is unavailable because he is on a business trip to Northeastern Pennsylvania. Jones updates her research on Pure Energy and then recommends the stock to Lisa Wong, CFA, a portfolio manager who purchases significant positions in client accounts. The following week, Pure Energy announces that they have entered into an agreement to be purchased by Alpine for a significant premium. Has either Jones or Wong *most likely* violated Standards with regards to the integrity of capital markets?

A No.

B Yes, Jones' recommendation is based on insider information.

C Yes, both Jones and Wong have acted on insider information.

A is correct because Jones has used the mosaic theory to combine non-material, non-public information with material public information.

B is incorrect because if taken in isolation, the information Jones received on the location of a business trip would be considered non-material.

C is incorrect because if taken in isolation, the information Jones received on the location of a business trip would be considered non-material.

Guidance for Standards I–VII
LOS a, b
Standard II(A)–Material Nonpublic Information

2 Ian O'Sullivan, CFA, is the owner and sole employee of two companies, a public relations firm and a financial research firm. The public relations firm entered into a contract with Mallory Enterprises to provide public relations services, with O'Sullivan receiving 40,000 shares of Mallory stock in payment for his services. Over the next 10 days, the public relations firm issued several press releases that discussed Mallory's excellent growth prospects. O'Sullivan, through his financial research firm, also published a research report recommending Mallory stock as a "buy." According to the CFA Institute Standards of Professional Conduct, O'Sullivan is *most likely* required to disclose his ownership of Mallory stock in the:

A press releases only.

B research report only.

C both the press release and the research report.

C is correct because members should disclose all matters that reasonably could be expected to impair the member's objectivity [Standard I(B), Standard VI(A)].

A is incorrect because both the press release and the research report should disclose any potential conflict of interest.

B is incorrect because both the press release and the research report should disclose any potential conflict of interest.

Guidance for Standards I–VII
LOS a
Standard I(B)–Independence and Objectivity, Standard VI(A)–Disclosure of Conflicts

3 Adira Badawi, CFA, who owns a research and consulting company, is an independent board member of a leading cement manufacturer in a small local market. Because of Badawi's expertise in the cement industry, a foreign cement manufacturer looking to enter the local market has hired him to undertake a feasibility study. Under what circumstances can Badawi *most likely* undertake the assignment without violating the CFA Institute Code of Ethics and Standards of Professional Conduct? If he:

A makes full disclosure to both companies.

B receives written permission from the local company.

C signs confidentiality agreements with both companies.

A is correct because making full and fair disclosure of all matters that could reasonably be expected to impair one's independence and objectivity or interfere with respective duties to one's clients is required by Standard VI(A)–Disclosure of Conflicts of the CFA Institute Code of Ethics and Standards of Professional Conduct.

B is incorrect because written permission from both parties would be needed to provide full and fair disclosure of all matters that could reasonably be expected to impair their independence and objectivity or interfere with respective duties to their clients. The requirement to disclose under Standard VI does not mandate that this be in writing. In fact, members and candidates have the responsibility of determining how often, in what manner, and in what particular circumstances the disclosure must be made [Standard IV(B)–Additional Compensation Arrangements requires written consent].

C is incorrect because the signing of confidentiality agreements does not necessarily provide full and fair disclosure of all matters that could reasonably be expected to impair Badawi's independence and objectivity or interfere with respective duties to his clients as required by Standard VI. Confidentiality agreements could actually restrict the disclosure of information that would provide fair disclosure.

Guidance for Standards I–VII
LOS c
Standard VI(A)–Disclosure of Conflicts

4 In order to provide investors with a more comprehensive view of a firm's performance, the current GIPS standards includes new provisions related to:

A various measures of risk.

B all aspects of performance measurement.

C the unique characteristics of each asset class.

A is correct. Historically, the GIPS standards focused primarily on returns. In the spirit of fair representation and full disclosure, and in order to provide investors with a more comprehensive view of a firm's performance, the current GIPS standards includes new provisions related to risk.

B is incorrect, the GIPS standards do not address all aspects of performance measurement.

C is incorrect, the GIPS standards do no cover the unique characteristics of each asset class.

The GIPS Standards
LOS a

5 Darden Crux, CFA, a portfolio manager at SWIFT Asset Management Ltd. (SWIFT), calls a friend to join him for dinner. The friend, a financial analyst at Cyber Kinetics (CK) declines the invitation and explains she is performing due diligence on Orca Electronics, a company CK is about to acquire. After the phone call, Crux searches the Internet for any news of the acquisition but finds nothing. Upon verifying that Orca is on SWIFT's approved stock list, Crux purchases Orca's common stock and call options for selective SWIFT clients. Two weeks later, CK announces its intention to acquire Orca. The next day, Crux sells all of the Orca securities, giving the fund a profit of $3 million. What action should Crux *most likely* take to avoid violating any CFA Institute Standards of Professional Conduct?

A Refuse to trade based on the information.

B Purchase the stock and call options for all clients.

C Trade only after analyzing the stock diligently and thoroughly.

A is correct as members/candidates who possess material nonpublic information that could affect the value of an investment should not act or cause others to act on the information. Crux traded on the material information that Orca is about to be acquired by Cyber Kinetics. The information is nonpublic because it is not publicly available, which was verified when Crux researched Orca on the Internet and found nothing about the acquisition [Standard II(A)].

B is incorrect because the information is material and nonpublic and should not be traded.

C is incorrect because the information is material and nonpublic and should not be traded.

Guidance for Standards I–VII
LOS b
Standard II(A)–Material Nonpublic Information

6 Alexander Newton, CFA, is the chief compliance officer for Mills Investment Limited. Newton institutes a new policy requiring the pro rata distribution of new security issues to all established discretionary accounts for which the new issues are appropriate. The policy also provides for the exclusion of newly established discretionary accounts from the distribution until they have reached their one-month anniversary date. This policy is disclosed to all existing and potential clients. Did Newton *most likely* violate any CFA Institute Standards of Professional Conduct?

A Yes.

B No, because this allocation policy is not inequitable under the Standards.

C No, because this policy has been adequately disclosed to all existing and potential clients.

A is correct because under Standard III(B)–Fair Dealing, members and candidates should disclose to clients and prospective clients how they select accounts to participate in an order and how they determine the amount of securities each account will buy or sell. Trade allocation procedures must be fair and equitable, and disclosure of inequitable allocation methods does not relieve the member or candidate of this obligation. All discretionary accounts should be treated in the same manner. Treating newer accounts differently would be considered inequitable regardless if this policy is disclosed.

B is incorrect because making new accounts wait for an arbitrary anniversary date before participating is not fair or equitable.

C is incorrect because under Standard III(B) inequitable allocation methods cannot be disclosed away.

Guidance for Standards I–VII
LOS a
Standard III(B)–Fair Dealing

7 Which of the following distinct entities can *least likely* claim compliance with the Global Investment Performance Standards (GIPS)?

A A multi-national financial services holding company

B An investment management division of a regional commercial bank

C A locally incorporated subsidiary undertaking investment management services

A is correct because the Global Investment Performance Standards require that firms be defined as an investment firm, subsidiary, or division held out to clients or prospective clients as a distinct business entity (0.A.12). A multi-national financial services holding company is unlikely to be solely operating as an investment firm, and the scope of the business could also make it more difficult to claim compliance on a firm-wide basis.

B is incorrect because an investment management division or a regional commercial bank could fit the definition of a firm as defined in 0.A.12.

C is incorrect because an investment management division of a regional commercial bank could fit the definition of a firm as defined in 0.A.12.

Global Investment Performance Standards (GIPS)
LOS b
GIPS Requirement 0.A.12

8 Prudence Charmaine, a CFA charterholder, was recently accused in writing of cheating on a professional accounting exam. She denied cheating and successfully defended herself against the allegation. As part of her defense and as evidence of her character, Charmaine stated that she is a CFA charterholder and upholds the CFA Institute Code of Ethics and Standards of Professional Conduct. On her next annual Professional Conduct Statement, Charmaine does not report this allegation to CFA Institute. Did Charmaine *most likely* violate the CFA Institute Code of Ethics or Standards of Professional Conduct?

A No

B Yes, she improperly used the CFA Institute Code and Standards to defend herself.

C Yes, she did not report the allegation on her annual Professional Conduct Statement.

C is correct because Charmaine should have reported the cheating allegation when making her annual Professional Conduct Statement. Even though she successfully defended herself against the charges and the charges were dropped, she has a responsibility to report the written complaint involving her integrity. The Code of Ethics requires CFA charterholders to practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

A is incorrect because Charmaine should have reported the cheating charges and the subsequent successful defense when making her annual Professional Conduct Statement. Even though she successfully defended herself against the charges, she has a responsibility to report the written complaint involving her integrity. The Code of Ethics requires CFA charterholders to practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

B is incorrect because it is not apparent that Charmaine violated Standard VII(B)–Reference to CFA Institute, the CFA Designation, and the CFA Program. Charmaine was correct in stating she is required to abide by the CFA Code and Standards.

CFA institute Code of Ethics and Standards of Professional Conduct
LOS c

9 Which of the following is **not** part of the nine major sections of the GIPS standards?

A Performance Fees

B Disclosure

C Input Data

A is correct. The nine major sections of the GIPS standards do not include performance fees. The nine major sections are fundamentals of compliance, input data, calculation methodology, composite construction, disclosure, presentation and reporting, real estate, private equity, and wrap fee/separately managed account portfolios.

B is incorrect because input data is one of the nine major sections of the GIPS standards.

C is incorrect because disclosure is one of the nine major sections of the GIPS standards.

The GIPS Standards
LOS d

10 Which of the following statements related to requirements for the CFA Institute Standards of Professional Conduct Standard V(B)–Communication with Clients and Prospective Clients is ***least likely*** accurate? The standard requires members and candidates to:

A divulge the number of investment related personnel responsible for external communication.

B disclose the basic format and general principles of the investment process.

C distinguish between fact and opinion in the presentation of investment analysis and recommendations.

A is correct. The CFA Institute Standards of Professional Conduct Standard V(B)–Communication with Clients and Prospective Clients does not limit the type or number of staff responsible for external communication.

B is incorrect because disclosure of the basic format and general principles of the investment process is a requirement.

C is incorrect because distinguishing between fact and opinion in the presentation of investment analysis and recommendations is a requirement.

Code of Ethics and Standards of Professional Practice
LOS b

11 Abdul Naib, CFA, was recently asked by his employer to submit an updated document providing the history of his employment and qualifications. The existing document on file was submitted when he was hired five years ago. His employer notices that the updated version shows that Naib obtained his Masters of Business Administration (MBA) two years ago, while the earlier version indicated he had already obtained his MBA. As the position Naib was hired for required a minimum qualification of an MBA, Naib is asked to explain the discrepancy. He justifies his actions by stating, "I knew you wouldn't hire me if I didn't have an MBA degree but I already had my CFA designation. Knowing you required an MBA, I went back to school on a part-time basis after I was hired to obtain it. I graduated at the top of my class, but this shouldn't come as any surprise, as you have seen evidence I passed all of my CFA exams on the first attempt." Did Naib most likely violate the CFA Institute Standards of Practice?

A No.

B Yes, with regard to Misconduct.

C Yes, with regard to Reference to the CFA Designation.

B is correct because Naib knowingly misrepresented his qualifications by stating he had obtained an MBA degree at the time of his hire when in fact he had not. This reflects adversely on his professional integrity, violating Standard I(D)–Misconduct. Stating that he passed his CFA exams in three consecutive years is not a violation of Standard VII(B)–Reference to CFA Institute, the CFA Designation, and the CFA Program if it is factual. There is no evidence given to indicate he did not pass as claimed.

A is incorrect because Naib knowingly misrepresented himself by stating he had obtained an MBA degree when in fact he had not. This reflects adversely on his professional integrity, violating Standard I(D)–Misconduct.

C is incorrect because stating he passed his CFA exams consecutively is not a violation of Standard VII(B)–Reference to CFA Institute, the CFA Designation, and the CFA Program. There is no evidence given to indicate he did not pass as claimed.

Guidance for Standards I–VII
LOS b

12 Victoria Christchurch, CFA, is a management consultant currently working with a financial services firm interested in curtailing its high staff turnover, particularly amongst CFA charterholders. In recent months, the company lost 5 of its 10 most senior managers, all of whom have cited systemic unethical business practices as the reason for their leaving. To curtail staff turnover by encouraging ethical behavior, it would be *least* appropriate for Christchurch to recommend the company to do which of the following?

A Implement a whistleblowing policy.

B Encourage staff retention with increased benefits.

C Create, implement, and monitor a corporate code of ethics.

B is correct because the offering of increased benefits to encourage staff retention would not necessarily stop the unethical behavior causing staff turnover and would effectively be asking the ethical employees to ignore the unethical behavior, thus being complicit in the behavior. Under Standard I(A)–Knowledge of the Law, CFA charterholders and candidates must disassociate themselves from unethical behavior. As the unethical business practices are seen as systemic, it would likely require them to leave the firm. Implementing a whistleblowing policy and adopting a corporate code of ethics would likely help to build a foundation of strong ethical behavior.

A is incorrect as introducing a whistleblowing policy would likely help to build a foundation of strong ethical behavior

C is incorrect as implementing a corporate code of ethics would likely help to build a foundation of strong ethical behavior.

Guidance for Standards I–VII
LOS c

13 Millicent Plain has just finished taking Level II of the CFA examination. Upon leaving the examination site, she meets with four Level III candidates who also just sat for their exams. Curious about their examination experience, Plain asks the candidates how difficult the Level III exam was and how they did on it. The candidates say the essay portion of the examination was much harder than they had expected and they were not able to complete all questions as a result. The candidates go on to tell Plain about broad topic areas that were tested and complain about specific formulas they had memorized that did not appear on the exam. The Level III candidates *least likely* violated the CFA Institute Standards of Professional Conduct by discussing:

A specific formulas.

B broad topic areas.

C the examination essays.

C is correct because discussing the level of difficulty of the essay portion of the examination did not violate Standard VII(A)–Conduct as Members and Candidates in the CFA Program. Standard VII(A) and the Candidate Pledge were violated by candidates revealing broad topical areas and formulas tested or not tested on the exam.

A is incorrect as Standard VII(A)–Conduct as Members and Candidates in the CFA Program and the Candidate Pledge was violated by candidates revealing specific formulas

B is incorrect as Standard VII(A)–Conduct as Members and Candidates in the CFA Program and the Candidate Pledge was violated by candidates revealing portions of the CBOK covered on the exam and areas not covered

Guidance for Standards I–VII
LOS b
Standard VII(A)–Conduct as Participants in CFA Institute Programs

14 Carlos Cruz, CFA, is one of two founders of an equity hedge fund. Cruz manages the fund's assets while the other co-founder, Brian Burkeman, CFA, is responsible for fund sales and marketing. Cruz notices the most recent sales material used by Burkeman indicates that assets under management are listed at a higher value than the current market value. Burkeman justifies the discrepancy by stating recent market declines account for the difference. In order to comply with the CFA Institute Standards of Professional Conduct, Cruz should *least likely* take which of the following actions?

A Correct the asset information and provide updates to prospective clients.

B Report the discrepancy to the CFA Institute Professional Conduct Program.

C Provide a disclaimer within marketing material indicating prices are as of a specific date.

B is correct because a violation of Standard I(A)–Knowledge of the Law is likely to occur unless the asset base information is corrected. Cruz has yet to violate any CFA Institute Standards so he need not report a violation. If Cruz does not take action he will be in violation of the Standards and at that point he would need to report this violation because Standard I(A) applies as the member should know his conduct may contribute to a violation of applicable laws, rules, regulations, or the Code and Standards related to the inaccurate sales materials. Cruz should seek to have the information corrected and accurate information provided to prospective clients. It may also be prudent to seek the advice of legal counsel.

A is incorrect as Cruz should seek to have the information corrected and accurate information provided to prospective clients.

C is incorrect as providing a disclaimer within the marketing material concerning the date of the market prices would be a prudent step since market prices are likely to change frequently from when the material was published.

Guidance for Standards I–VII
LOS c
Standard I(A)–Knowledge of the Law

15 When a client asks her how she makes investment decisions, Petra Vogler, CFA, tells the client she uses mosaic theory. According to Vogler, the theory involves analyzing public and nonmaterial nonpublic information including the evaluation of statements made to her by company insiders in one-on-one meetings where management discusses new earnings projections not known to the public. Vogler also gathers general industry information from industry experts she has contacted. Vogler *most likely* violates the CFA Institute Standards of Professional Conduct because of her use of:

A industry expert information.

B one-on-one meeting information.

C nonmaterial nonpublic information.

B is correct because a violation of Standard II(A)–Material Nonpublic Information is likely to occur when using information that is selectively disclosed by corporations to a small group of investors, analysts, or other market participants. Earnings estimates given in a one-on-one meeting would likely be considered material and nonpublic information. Information made available to analysts remains nonpublic until it is made available to investors in general. Under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions. Additionally, it is acceptable to use nonmaterial nonpublic information in her analysis, and this use is not a violation of Standard II(A)–Material Nonpublic Information.

A is incorrect because under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions.

C is incorrect because it is acceptable to use nonmaterial nonpublic information in her analysis, and this use is not a violation of Standard II(A)–Material Nonpublic Information.

Guidance for Standards I–VII
LOS b
Standard II(A)–Material Nonpublic Information

16 Noor Hussein, CFA, runs a financial advisory business, specializing in retirement planning and investments. One of her clients asks her to advise the firm's pension fund trustees on available investments in the market including Islamic products. On the day prior to the meeting, Hussein spends an hour familiarizing herself with Islamic investment products and getting updates on local market conditions. The next day, she recommends Islamic investment products to the trustees based on her research and her expertise in retirement planning and investments. The trustees subsequently incorporate Islamic products into their investment allocation. Did Hussein's basis for the recommendation *most likely* comply with the CFA Code of Ethics?

A Yes.

B No, with regard to Misconduct.

C No, with regard to Diligence and Reasonable Basis.

C is correct because Hussein did not likely act with competence and diligence as required by Standard V(A). One half day of preparation with regard to Islamic investment products would not likely be considered sufficient to give investment advice to pension plan trustees. Misconduct was not violated by Hussein stating she is an expert in retirement planning and investments because this is the area she specializes in.

A is incorrect because Hussein did not likely act with competence and diligence as required by Standard V(A). One half day of preparation with regard to Islamic investment products would not likely be considered sufficient to give investment advice to pension plan trustees.

B is incorrect because it is not likely she violated Standard I(D)–Misconduct, i.e., conduct involving dishonesty, fraud, and/or deceit by stating she is an expert in retirement planning and investments.

Guidance for Standards I–VII
LOS b
Standard V(A)–Diligence and Reasonable Basis

17 Dimitri Kuznetsov, CFA, is a portfolio manager and holds shares of Barnikoff Limited and Matric Ventures in all client portfolios. Both companies have upcoming annual general meetings scheduled for the same day. The management of Barnikoff proposes to change its financial year-end from September to December, while Matric Ventures proposes to enter into a high-risk venture. The proxy voting policy clause in all client investment management agreements managed by Kuznetsov states, "When voting proxies provides a cost benefit to the client, the manager must vote a proxy." With regard to the proxy votes for Matric and Barnikoff, Kuznetsov would *least likely* violate CFA Institute Standard III(A)–Loyalty, Prudence, and Care if he votes:

A with management.

B only the Matric proxy.

C only the Barnikoff proxy.

B is correct because Standard III(A)–Loyalty, Prudence, and Care states that it is a member or candidate's duty to vote proxies on behalf of clients in an informed and responsible manner. However, if a cost–benefit analysis shows voting all proxies may not benefit the client, voting all proxies may not be necessary. The member or candidate is responsible for informing all clients if this is the policy of the fund manager. The member or candidate must take steps to disclose this proxy voting policy to clients. Voting the Barnikoff proxy does not appear to offer a benefit because the issue is not of a critical nature, but voting the proxy for Matric involves a material issue and is a benefit that should be voted on.

A is incorrect because Standard III(A)–Loyalty, Prudence, and Care states that it is a member or candidate's duty to vote proxies on behalf of clients in an informed and responsible manner. A manager must not blindly vote with management without first considering the impact of the issue at hand and its benefit to the client.

C is incorrect because while Standard III(A)–Loyalty, Prudence, and Care states that it is a member or candidate's duty to vote proxies on behalf of clients in an informed and responsible manner, if a cost–benefit analysis shows voting all proxies may not benefit the client, voting all proxies may not be necessary.

Guidance for Standards I–VII
LOS b
Standard III(A)–Loyalty, Prudence, and Care

18 Merchant Capital Partners, a regional investment bank, acts as a market maker for Vital Link Health Services and other small firms listed on an over-the-counter exchange. For those shares for whom Merchant acts as market maker, it trades for its own book as well as engaging in risk arbitrage trading. Merchant allows staff members to trade in shares once clients and the company have traded. Merchant recently obtained material nonpublic information regarding Vital's planned reverse takeover of a publicly listed competitor. In order to be in compliance with the CFA Institute Code and Standards, which type of trading in Vital shares should Merchant *least likely* suspend?

A Personal

B Risk arbitrage

C Passive proprietary

C is correct because according to Standard II(A)–Material Nonpublic Information, Recommended Procedures for Compliance, if Merchant stopped market making, a form of proprietary trading, due to being in possession of material nonpublic information, it could tip off investors that Vital is likely to be making a major announcement in the near future. This would be counterproductive to the goals of maintaining the confidentiality of information and providing market liquidity. The Standard recommends that market makers remain passive when in possession of material nonpublic information. The Standard also requires personal trading to be suspended when in possession of material nonpublic information, and it is prudent to suspend arbitrage trading to prevent profits from insider trading.

A is incorrect because when in possession of material nonpublic information, Standard II(A)–Material Nonpublic Information requires personal trading to be suspended.

B is incorrect because when in possession of material nonpublic information, according to Standard II(A)–Material Nonpublic Information, it is prudent to suspend arbitrage trading to prevent profits from insider trading.

Guidance for Standards I–VII
LOS c
Standard II(A)–Material Nonpublic Information

19 Over the past four years, a portfolio experienced returns of –8%, 4%, 17%, and –12%. The geometric mean return of the portfolio over the four-year period is *closest* to:

A 0.25%.

B –0.37%.

C 0.99%.

B is correct. Add one to each of the given returns, then multiply them together and take the fourth root of the resulting product. $0.92 \times 1.04 \times 1.17 \times 0.88 = 0.985121$; 0.985121 raised to the 0.25 power is 0.996259. Subtracting one and multiplying by 100 gives the correct geometric mean return: $[(0.92 \times 1.04 \times 1.17 \times 0.88)^{0.25} - 1] \times 100 = -0.37\%$.

A is incorrect because it is the arithmetic mean of the four numbers.

C is incorrect because it is the solution to: $(0.92 \times 1.04 \times 1.17 \times 0.88) = 0.99$ (rounded).

Statistical Concepts and Market Returns
LOS e
Section 5.4.2

20 When an investigator wants to test whether a particular parameter is greater than a specific value, the null and alternative hypothesis are *best* defined as:

A H_0: $\theta \leq \theta_0$ versus H_a: $\theta > \theta_0$.

B H_0: $\theta \geq \theta_0$ versus H_a: $\theta < \theta0$.

C H_0: $\theta = \theta_0$ versus H_a: $\theta \neq \theta$

A is correct. A positive "hoped for" condition means that the null will be rejected (and the alternative accepted) only if the evidence indicates that the population parameter is greater than θ_0. Thus, H_0: $\theta \leq \theta_0$ versus H_a: $\theta > \theta_0$ is the correct statement of the null and alternative hypotheses, respectively.

B is incorrect; it can only discern that a parameter is possibly lesser than a value.

C is incorrect; it can only discern that a parameter is not equal to a value.

Hypothesis Testing
LOS a, b
Section 2

21 An analyst has established the following prior probabilities regarding a company's next quarter's earnings per share (EPS) exceeding, equaling, or being below the consensus estimate.

	Prior Probabilities
EPS exceed consensus	25%
EPS equal consensus	55%
EPS are less than consensus	20%

Several days before releasing its earnings statement, the company announces a cut in its dividend. Given this new information, the analyst revises his opinion regarding the likelihood that the company will have EPS below the consensus estimate. He estimates the likelihood the company will cut the dividend, given that EPS exceeds/meets/falls below consensus, as reported below.

	Probabilities the Company Cuts Dividends, Conditional on EPS Exceeding/Equaling/Falling below Consensus
P(Cut div $\vert$ EPS exceed)	5%
P(Cut div $\vert$ EPS equal)	10%
P(Cut div $\vert$ EPS below)	85%

The analyst thus determines that the unconditional probability for a cut in the dividend, P(Cut div), is equal to 23.75%. Using Bayes' formula, the updated (posterior) probability that the company's EPS are below the consensus is *closest* to:

A 85%.

B 72%.

C 20%.

B is correct. Bayes' Formula:

Updated probability of event given the new information

$$= \frac{\text{Probability of the new information given event}}{\text{Unconditional probability of the new information}} \times \text{Prior probability of event}$$

where

Updated probability of event given the new information: *P*(EPS below | Cut div);

Probability of the new information given event: *P*(Cut div | EPS below) = 85%;

Unconditional probably of the new information: *P*(Cut div) = 23.75%;

Prior probability of event: *P*(EPS below) = 20%.

Therefore, the probability of EPS falling below the consensus is updated as:

$$P(\text{EPS below} \mid \text{Cut div}) = [P(\text{Cut div} \mid \text{EPS below})/P(\text{Cut div})] \times P(\text{EPS below})$$
$$= (0.85/0.2375) \times 0.20 = 0.71579 \sim \mathbf{72\%}$$

B is incorrect. It is the given *P*(Cut div | EPS below).

C is incorrect. It simply multiplies the unconditional probability for a cut in the dividend with the conditional probability of a cut in the dividend given that EPS falls below consensus: *P*(Cut div) × *P*(Cut div | EPS below) = 0.2375 × 0.85 = 20.188.%

Probability Concepts
LOS h, n
Sections 2, 4.1

22 Consider the investment in the following table:

Start of Year 1	One share purchased at $100
End of Year 1	$5.00 dividend/share paid and one additional share purchased at $125
End of Year 2	$5.00 dividend/share paid and both shares sold for $140 per share

Assuming dividends are not reinvested, compared with the time-weighted return, the money-weighted return is:

A lower.

B the same.

C higher.

A is correct. The following table represents cash flows of the investment:

Year	Contribution	Start-of-Year Value after Contribution	End-of-Year Dividend	End-of-Year Value after Dividend
1	1 × $100	1 × $100 = $100	1 × $5 = $5	$125
2	1 × $125	2 × $125 = $250	2 × $5 = $10	(2 × 140) + 10 = $290

The time-weighted rate of return (TWR) on this investment is found by taking the geometric mean of the two holding period returns (HPRs):

$$\text{TWR} = [(1 + \text{HPR}_{\text{Year 1}}) \times (1 + \text{HPR}_{\text{Year 2}})]^{1/2} - 1$$

where

$$\begin{aligned} HPR_{Year\ 1} &= (\$125 - \$100 + \$5)/\$100 = 30.0\% \\ HPR_{Year\ 2} &= (\$280 - \$250 + \$10)/\$250 = 16.0\% \\ TWR &= [(1 + 0.30) \times (1 + 0.16)]^{1/2} - 1 = 22.80\% \end{aligned}$$

The money-weighted rate of return (MWR) is the internal rate of return (IRR) of the cash flows associated with the investment:

$$0 = -100 + \frac{(-125 + 5)}{(1 + r_1)} + \frac{(280 + 10)}{(1 + r_2)}, \text{ where } r = \text{MWR}.$$

Using the cash flow (CF) function of a financial calculator:

$CF_0 = -100$, $CF_1 = (-125 + 5)$, $CF_2 = (280 + 10)$, and solving for IRR: MWR or IRR = 20.55%.

The difference between the TWR and MWR of this investment = 22.80% – 20.55% = 2.25%, or 225 bps, with MWR being lower than TWR.

B is incorrect. The difference between MWR and TWR is either higher or lower, and could hardly be equal.

C is incorrect, as per the calculation above.

Discounted Cash Flow Applications
LOS d
Sections 3.1 and 3.2

23 An analyst collects data relating to five commonly used measures of financial leverage and interest coverage for a randomly chosen sample of 300 firms. The data come from those firms' fiscal year 2013 annual reports. These data are *best* characterized as:

A longitudinal.

B cross sectional.

C time series.

B is correct. Data on some characteristics of companies at a single point in time are cross-sectional data.

A is incorrect. The data are not longitudinal data. Longitudinal data are observations on characteristic(s) of the same observational unit through time.

C is incorrect. The data are not time-series data. Time-series data are observations of a variable over time

Sampling and Estimation
LOS d
Section 2.3

24 The following information applies to a sample:

- The point estimate of the population mean is 12.5.
- The *t*-statistic ($t_{\alpha/2}$) used in calculating the 90% confidence interval is 1.67.
- The sample size is 64.
- The sample standard deviation is 5.

The 90% confidence interval for the population mean is *closest* to:

A 11.98 to 13.02.

B 12.37 to 12.63.

C 11.46 to 13.54.

C is correct. The confidence interval for the population mean is calculated as:

$$\bar{X} \pm t_{\alpha/2}\left(s/\sqrt{n}\right)$$

where $\bar{X}$ is the mean of the sample (12.5), $t_{\alpha/2}$ is the appropriate *t*-statistic (1.67), *s* is the sample standard deviation (5), and *n* is the sample size (64). In this problem, the confidence interval is 12.5 ± 1.67 × $\left(5/\sqrt{64}\right)$ = 12.5 ± 1.04375 = 11.45625 to 13.54375, rounded to **11.46** to **13.54**.

A is incorrect. The mistake is to divide 1.67 by 2 rather than use the value as given: 12.5 ± (1.67/2) × $\left(5/\sqrt{64}\right)$ = 12.5 ± 0.52188 = 11.97812 to 13.02188, rounded to 11.98 to 13.02.

B is incorrect. The mistake is using 64 instead of the square root of 64 in the calculation: 12.5 ± 1.67 × (5/64) = 12.5 ± 0.13047 = 12.36953 to 12.63047, rounded to 12.37 to 12.63.

Sampling and Estimation
LOS j
Section 4.2

25 Consider a two-tailed test of the hypothesis that the population mean is zero. The sample has 50 observations. The population is normally distributed with a known variance.

***t*-Test rejection level**			
Degrees of freedom	***p* = 0.10**	***p* = 0.05**	***p* = 0.025**
49	1.299	1.677	2.010
50	1.299	1.676	2.009
***z*-Test rejection level**	**α = 0.10**	**α = 0.05**	**α = 0.025**
	1.645	1.960	2.330

At a 0.05 significance level, the rejection points are *most likely* at:

A –2.009 and 2.009.

B –2.010 and 2.010.

C –1.960 and 1.960.

C is correct. The appropriate test statistic is a *z*-statistic because the sample comes from a normal distributed population with known variance. A *z*-test does not use degrees of freedom. This test is two-sided at the 0.05 significance level, and the rejection point conditions are $z > 1.960$ and $z < -1.960$.

B is incorrect. As explained in choice C.

A is incorrect. As explained in choice C.

Hypothesis Testing
LOS g
Section 3

26 A risk manager would like to calculate the coefficient of variation of a portfolio. The following table reports the annual returns of the portfolio and of the risk-free rate over the most recent five years:

Year	Portfolio Return	Risk-Free Rate
1	4.0%	2.0%
2	–1.0%	1.5%
3	7.0%	1.0%
4	11.0%	1.0%
5	2.0%	0.5%

The coefficient of variation of the portfolio is *closest* to:

A 1.00.

B 0.74.

C 0.90.

A is correct. First calculate the sample mean return as follows:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

where

n = the number of observations in the sample

i = the index for the year

X_i = is the return in year i

$$\bar{X} = \frac{(4.0\% - 1.0\% + 7.0\% + 11.0\% + 2.0\%)}{5}$$

$$= \frac{23.0\%}{5}$$

$$= 4.6\%$$

Then calculate the sample standard deviation with the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

Year	$(X_i - \bar{X})^2$
1	$(4.0\% - 4.6\%)^2 = 0.00004$
2	$(-1.0\% - 4.6\%)^2 = 0.00314$
3	$(7.0\% - 4.6\%)^2 = 0.00058$
4	$(11.0\% - 4.6\%)^2 = 0.00410$
5	$(2.0\% - 4.6\%)^2 = 0.00068$

$$s = \sqrt{\frac{0.00004 + 0.00314 + 0.00058 + 0.00410 + 0.00068}{5-1}} = 4.62\%$$

The coefficient of variation (CV) is calculated with the following formula:

$$\mathrm{CV} = \frac{s}{\overline{X}} = \frac{4.62\%}{4.6\%} = 1.0$$

B is incorrect. It is the Sharpe ratio, not the coefficient of variation. First calculate the mean annual risk-free return over the five years:

$$\overline{R}_F = \frac{2.0\% + 1.5\% + 1.0\% + 1.0\% + 0.5\%}{5} = 1.2\%$$

Then calculate the Sharpe ratio with the following formula:

$$S_h = \frac{\overline{R}_p - \overline{R}_F}{s_p} = \frac{4.6\% - 1.2\%}{4.62\%} = 0.74$$

C is incorrect. In the formula for the standard deviation, it uses *n* instead of "*n* – 1" in the denominator:

$$s = \sqrt{\frac{0.00004 + 0.00314 + 0.00058 + 0.00410 + 0.00068}{5}} = 4.13\%$$

Then, calculate the CV:

$$\mathrm{CV} = \frac{4.13\%}{4.6\%} = 0.90$$

Statistical Concepts and Market Returns
LOS e, g, i
Sections 5.1.2, 7.4.2, 7.7

27 An economist states that the probability of having the gross domestic product (GDP) of a country higher than 3% is 0.20. What are the odds against a GDP higher than 3%?

A 5 to 1

B 6 to 1

C 4 to 1

C is correct. Given the probability of an event, $P(E)$, the odds against that event are: $[1 - P(E]/P(E)$. By using the input of the problem, Odds against $E = (1 - 0.2)/0.2 = 4$. This means that given the probability stated by the economist, the odds against a GDP above 3% are 4 to 1.

A is incorrect. It uses a wrong calculation: (1/0.20) = 5. Thus, 5 to 1.

B is incorrect. It uses a wrong formula: (1 + 0.2)/0.2 = 6. Thus, 6 to 1.

Probability Concepts
LOS c
Section 2

28 The joint probability of returns for securities A and B are as follows:

Joint Probability Function of Security A and Security B Returns (Entries Are Joint Probabilities)		
	Return on Security B = 30%	**Return on Security B = 20%**
Return on Security A = 25%	0.60	0
Return on Security A = 20%	0	0.40

The covariance of the returns between Securities A and B is *closest* to:

A 12.

B 14.

C 13.

A is correct. First calculate the expected returns on securities A and B with the formula:

$$E(X) = \sum_{i=1}^{n} P(X_i)X_i$$

Expected return on security A = 0.6 × 25% + 0.4 × 20% = 15% + 8% = 23%
Expected return on security B = 0.6 × 30% + 0.4 × 20% = 18% + 8% = 26%
Then calculate the covariance of returns between securities A and B with the formula:

$$\text{Cov}(R_A,R_B) = \sum_i \sum_j P\left(R_{A,i},R_{B,j}\right)\left(R_{A,i} - ER_A\right)\left(R_{B,j} - ER_B\right)$$

where

R_A and R_B = the returns on securities A and B, respectively
P = the joint probability
ER_A and ER_B = the expected returns of securities A and B, respectively
i and *j* = the line and column of the joint probability function table above

$$\begin{aligned}\text{Cov}\left(R_A,R_B\right) &= 0.6\left[(25-23)(30-26)\right] + 0.4\left[(20-23)(20-26)\right]\\ &= 0.6\left[2\times 4\right] + 0.4\left[(-3)(-6)\right]\\ &= 0.6\times 8 + 0.4\times 18\\ &= 4.8 + 7.2\\ &= 12\end{aligned}$$

B is incorrect. In the covariance calculation, it uses the joint probabilities in the wrong positions:

$$\begin{aligned}\text{Cov}\left(R_A,R_B\right) &= 0.4\left[(25-23)(30-26)\right] + 0.6\left[(20-23)(20-26)\right]\\ &= 0.4\left[2\times 4\right] + 0.6\left[(-3)(-6)\right]\\ &= 0.4\times 8 + 0.6\times 18\\ &= 3.2 + 10.8\\ &= 14\end{aligned}$$

C is incorrect. In the covariance calculation, it uses a joint probability of 0.5:

$$\begin{aligned}\mathrm{Cov}(R_A, R_B) &= 0.5\big[(25-23)(30-26)\big] + 0.5\big[(20-23)(20-26)\big] \\ &= 0.5[2 \times 4] + 0.5\big[(-3)(-6)\big] \\ &= 0.5 \times 8 + 0.5 \times 18 \\ &= 4 + 9 \\ &= 13\end{aligned}$$

Probability Concepts
LOS m
Section 3

29 An investor deposits £2,000 into an account that pays 6% per annum compounded continuously. The value of the account at the end of four years is *closest* to:

A £2,854.

B £2,525.

C £2,542.

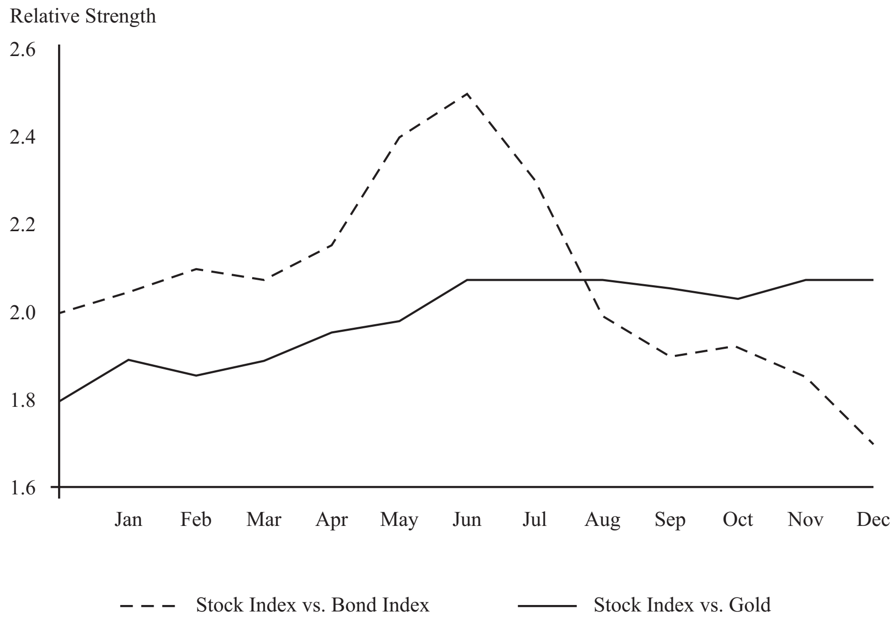
C is correct. The future value (FV) of a given lump sum, calculated using continuous compounding, is: FV = PVe^{rN} = 2,000 × $e^{0.06 \times 4}$ = £2,542.49 ~ £2,542.

A is incorrect. It is calculated as: ln(0.06 × 4) × 2,000 = –2,854.23 and ignores the negative sign of the logarithm.

B is incorrect. It uses annual compounding: 2,000 × $(1 + 0.06)^4$ = 2,524.95.

The Time Value of Money
LOS d, e
Section 3.2

30 The chart below depicts the relative strength lines for a stock index versus both a bond index and gold. In the month of June, it would be *most* appropriate for an analyst using intermarket analysis to move investments from:



A gold to stocks.

B stocks to bonds.

C bonds to gold.

B is correct. In intermarket analysis, technicians often look for inflection points in one market as a warning sign to start looking for a change in trend in a related market. To identify these intermarket relationships, a commonly used tool is relative strength analysis, which charts the price of one security divided by the price of another. In June, only the dashed line shows an inflection point (a top and reversal of trend) and that is an indication that the stock index started to weaken relative to the bond index. Therefore, it is a signal that the time had come to move investments from stocks to bonds.

A is incorrect. Starting from June, the continuous line is basically flat, indicating that neither stocks nor gold have a relative strength compared to each other (neutral performance).

C is incorrect. Given that from June, bonds indicate relative strength vs. stocks and that there is no relative strength between stocks and gold, it can be inferred that bonds have also a relative strength vs. gold. Therefore, it is not appropriate moving from bonds to gold.

Technical Analysis
LOS h, b
Sections 3.1.8, 5

31 A normally distributed random variable has a mean of 100 and a standard deviation of 12. The probability of observing a value greater than 82 is the cumulative distribution function (cdf) of the standard normal variable:

A $N(1.5)$.

B $N(-1.5)$.

C $1 - N(1.5)$.

A is correct. The standardized value of this normal distribution can be obtained using the formula $(X - \mu)/\sigma = (82 - 100)/12 = -1.5$. The cdf of $N(-1.5)$ provides the probability of a value less than or equal to 82.

B is incorrect because $1 - N(1.5) = N(-1.5)$ is the probability of a value in this distribution of less than 82.

C is incorrect because the probability of observing random draw less than 82 will be $P(Z < -1.5) = N(-1.5) = 1 - N(1.5)$.

Common Probability Distributions
LOS l
Section 3.2

32 The following statistical table is presented to an analyst

X* = *x	**Probability Function $P(x) = P(X = x)$**	**Cumulative Distribution Function $F(x) = P(X \leq x)$**
1	0.10	0.10
2	0.15	0.25
3	0.50	0.75
4	0.15	0.90
5	0.10	1.00

Which of the following is the *best* conclusion that can be drawn from the statistical table?

A The probability that the value of X will be equal to 4 is 0.90

B The probability that the value of X will be less than or equal to 5 but greater than 2 is 0.75.

C The random variable X is continuous.

B is correct. The probability that X is greater than 2 but less than or equal to 5 is: $P(2 < X \leq 5) = [P(X \leq 5) - P(X \leq 2)] = 1.00 - 0.25 = 0.75$.

A is incorrect because a random variable value of $X = 4$ has a 15% probability of occurring. The cumulative probability of 90% shows the probability of $X \leq 4$.

C is incorrect because the random variable has a countable number of possible values; the probability and cumulative distribution functions are therefore for a discrete random variable is therefore discrete.

Common Probability Distributions
LOS d
Section 2

33 To project the assets and liabilities of a pension plan using a number of different assumptions, the *most* appropriate method to employ is:

A Monte Carlo simulation because it can evaluate the effect of changes in assumptions.

B Monte Carlo simulation because it can provide more insights into cause and effect relationships than analytical methods.

C historical simulation because risk not represented in the time period observed can be integrated in the simulation.

A is correct. Monte Carlo simulation is better suited to hypothetical "what if" analysis than historical simulation. To model a pension plan's status under a number of different hypothetical assumptions affecting the assets and liabilities, Monte Carlo simulation is the most appropriate method

B is incorrect because Monte Carlo simulation is not an analytical method that can provide great insights into cause and effect relationships. It is a complement to analytical methods.

C is incorrect because historical simulation is not as well suited to hypothetical "what if" analysis as is Monte Carlo simulation. While historical simulation is easier to model because it relies on data from the past to predict future probability distributions, it is not as well suited for meeting the goal of projecting the plan's assets and liabilities under different assumptions.

Common Probability Distributions
LOS q
Section 4

34 If the quantity demanded of pears falls by 4% when the price of apples decreases by 3%, then apples and pears are *best* described as:

A complements.

B substitutes.

C inferior goods.

B is correct. The cross-price elasticity of demand is defined as the percentage change in quantity demanded divided by the percentage change in the price of a substitute or complement. If the cross-price elasticity of demand is positive, the goods are substitutes. In this case, the 4 percent decline in quantity of pears is divided by the 3 percent decline in the price of apples, which is a positive number:

$$-4/-3 = +1.33$$

A is incorrect because the cross-price elasticity of demand between pears and apples is positive. Complements have a negative cross-price elasticity of demand.

C is incorrect because inferior goods relates to income elasticity of demand, not cross-price elasticity.

Topics in Demand and Supply Analysis
LOS a
Section 2.4

35 According to the Solow neoclassical growth model, sustained long-term growth in potential GDP is *best* explained by:

A capital deepening investments.

B technological change.

C growth in labor supply.

B is correct. Growth in potential GDP equals growth in technology plus the weighted average growth rate of labor and capital. Because of diminishing marginal productivity (return), the only way to sustain long-term growth in potential GDP is through technological change or growth in total factor productivity.

A is incorrect. Growth in potential GDP equals growth in technology plus the weighted average growth rate of labor and capital. Because of diminishing marginal productivity (return), the only way to sustain long-term growth in potential GDP is through technological change or growth in total factor productivity.

C is incorrect. Growth in potential GDP equals growth in technology plus the weighted average growth rate of labor and capital. Because of diminishing marginal productivity (return), the only way to sustain long-term growth in potential GDP is through technological change or growth in total factor productivity.

Aggregate Output, Prices, and Economic Growth
LOS n, o
Section 4.1

36 A positive movement in a lagging indicator would *least likely* be used to:

A confirm that an expansion is currently underway.

B identify a past condition of the economy.

C identify an expected future economic upturn.

C is correct. A positive movement in a lagging indicator would most likely be used to confirm that an existing expansion is underway or has already occurred. Only a leading indicator would help identify or predict a future economic event.

A is incorrect. A positive move in a lagging indicator by itself is insufficient to indicate a positive expansion. However, confirmation would be required from positive changes in a coincident indicator to indicate expansion.

B is incorrect. A positive move in a lagging indicator is most likely identifying an upturn in economic activity that occurred in the past.

Understanding Business Cycles
LOS i
Section 5

37 Which of the following is the *least likely* outcome when a monopolist adopts first-degree price discrimination because of customers' differing demand elasticities?

A The output increases to the point at which price equals the marginal cost.

B The monopolist shares the total surplus with consumers.

C The price for a marginal unit decreases to less than the price for other units.

B is correct. In a monopoly, perfect price discrimination results in the total surplus being kept by the producer, the monopolist.

C is incorrect. When a monopolist adopts perfect price discrimination, the price for the marginal unit will be lower than average price.

A is incorrect. Under perfect price discrimination, output increases to the point where price equals marginal cost.

The Firm and Market Structures
LOS f
Section 6.4

38 If the scale of a single producer is small relative to the demand for an undifferentiated good, the market structure of the producer is *best* described as being:

A an oligopoly.

B monopolistic competition.

C perfect competition.

C is correct. Perfect competition involves the sale of a homogeneous product by many sellers; monopolistic competition may also involve many sellers, but its product is differentiated.

A is incorrect. Oligopolies involve only a few sellers.

B is incorrect. Firms in monopolistic competition sell differentiated products although there may be many of these firms.

The Firm and Market Structures
LOS a
Section 2.2

39 The following data apply to a country in its domestic currency units:

Consumer spending on goods and services	875,060	Government spending on goods and services	305,600
Business gross fixed investment	286,400	Government gross fixed investment	84,120
Change in inventories	–68,500	Capital consumption allowance	8,540
Transfer payments	9,300	Statistical discrepancy	–2,850
Exports	219,800	Imports	250,980

Using the expenditures approach, the country's GDP is *closest* to:

A 1,466,490.

B 1,451,500.

C 1,448,650.

C is correct. Using the expenditures approach:

GDP = Consumer spending on goods and services + Business gross fixed investment + Change in inventories + Government spending on goods and services + Government gross fixed investment + Exports – Imports + Statistical discrepancy

Consumer spending on goods and services	875,060
Business gross fixed investment	286,400

(continued)

Change in inventories	(68,500)
Government spending on goods and services	305,600
Government gross fixed investment	84,120
Exports	219,800
Imports	(250,980)
Statistical discrepancy	(2,850)
= Gross domestic product (GDP)	**1,448,650**

A is incorrect. It includes all items including transfer payments and capital consumption allowance, which are components of the income approach.

$$= 1{,}448{,}650 + 9{,}300 + 8{,}540 = 1{,}466{,}490$$

B is incorrect. It ignores statistical discrepancy.

$$= 1{,}448{,}650 + 2{,}850 = 1{,}451{,}500$$

Aggregate Output, Prices, and Economic Growth
LOS a
Sections 2.2, 2.3

40 Which of the following statements is *most* accurate based on the FX quotations in the table?

	Spot Rate	One-Year Forward Rate
USD/EUR	1.2952	1.3001

A The forward rate is trading at a discount to the spot rate by 0.0049 points.

B The euro is trading at a forward premium of 49 points.

C The US dollar is trading at a forward premium of 49 points.

B is correct. Forward premium = Forward rate − Spot rate = 1.3001 − 1.2952 = 0.0049. To convert to points, scale four decimal places—that is, multiply by 10,000 = 10,000 × 0.0049 = 49 points. Because the forward rate exceeds the spot rate for the base currency (euro), the euro is trading at a forward premium of 49 points.

C is incorrect. When the forward rate (1.3001) is higher than the spot rate (1.2952), the base currency (EUR) is said to be trading at a forward premium, not a discount, to the price currency (USD).

A is incorrect. When the forward rate (1.3001) is higher than the spot rate (1.2952), the forward points are positive, and the forward rate is said to be trading at a premium to the spot rate. The wrong number is derived from 1.3001 − 1.2952 = 0.0049. The USD/EUR is quoted to four decimal places, so it needs to be scaled up by four decimal places or multiplied by 10,000 to 49 points.

Currency Exchange Rates
LOS g
Section 3.3

41 Assume that a central bank has decided to lower interest rates in the economy. To carry out this policy, the central bank will *most likely*:

A increase required reserve requirements.

B buy securities.

C sell securities.

B is correct. In implementing monetary policy, central banks have three primary tools available to them: open market operations, setting the official policy rate, and reserve requirements. When the central bank purchases securities (open market operations), it increases the reserves held by private sector banks. These increased reserves lead to a reduction in interest rates on money market securities and, ultimately, to a reduction in other interest rates in the economy

A is incorrect. An increased reserve requirement reduces the money supply (as banks can lend out less) and leads to higher interest rates

C is incorrect. If the central bank sells securities, reserves fall and interest rates are likely to increase.

Monetary and Fiscal Policy
LOS h
Sections 2.3.2.1, 2.3.2.2

42 According to the concept of money neutrality, over the long term, the money supply is *least likely* to affect:

A inflation expectations.

B inflation.

C the real rate of interest.

C is correct. The concept of money neutrality implies that an increase in the money supply will leave real variables like output and employment unaffected. The real rate of interest will be unaffected by money supply changes but inflation and inflation expectations will be affected.

B is incorrect. According to money neutrality, real variables are not affected by changes in money supply but inflation will be.

A is incorrect. According to money neutrality, real variables are not affected by changes in money supply but inflation expectations can be.

Monetary and Fiscal Policy
LOS d, e
Sections 2.1.6, 2.1.7

43 Which of the following events would *most likely* have a positive impact on the GDP of Switzerland and the gross national product (GNP) of France?

A An industrial components manufacturer in France produces industrial components in France using Swiss workers.

B A Swiss company purchases machine tools from a manufacturing firm located in France.

C An accounting firm located in France provides accounting services to a manufacturer located in Switzerland.

C is correct. GDP measures the market value of all final goods and services produced by factors of production (such as labor and capital) located within a country, therefore Swiss GDP includes the accounting services provided within Switzerland. GNP measures the market value of all final goods and services produced by factors of production supplied by residents of a country, regardless of whether such production takes place within the country or outside of the country. Therefore, GNP includes the accounting services produced by a French citizen abroad.

A is incorrect. The production in France (outside of Switzerland) will not have a positive impact on Swiss GDP. Although it will be included in France's GDP (production in France by nationals and foreigners), it will not be included in France's GNP. GNP only includes value of goods and services produced by a country's residents.

B is incorrect. The purchase of French machine tools is an import which reduces Swiss GDP.

International Trade and Capital Flows
LOS a, h
Sections 2.1, 4.4

44 A member of the labor force quit her job last week and will begin a new job next week. During this interim period, for the purposes of calculating unemployment statistics, this person is *most likely* classified as:

A hidden unemployed.

B frictionally unemployed.

C voluntarily unemployed.

B is correct. Frictional unemployment is short term and transitory in nature; it includes people who are "between jobs," as in this case, and those who are not working because they are taking time to search for a job that better matches their skills, interests, and other preferences.

A is incorrect. The hidden unemployed encompass discouraged workers and underemployed people. Discouraged workers are those who have stopped looking for a job, and the underemployed are those who have a job but have qualifications to work at significantly higher paying jobs.

C is incorrect. Although this person likely left their old job voluntarily, frictionally unemployed is a better answer here because they do have a job waiting for them. The voluntarily unemployed are persons outside the labor force who might refuse an available vacancy if the pay rate is below their reservation wage or those who might have retired early.

Understanding Business Cycles
LOS d
Section 4.1

45 The primary goal of both monetary and fiscal policy focuses on balancing economic growth and:

A income distribution.

B inflation.

C employment.

B is correct. The goal of both monetary and fiscal policy is the creation of an economic environment characterized by positive, stable growth and low, stable inflation.

A is incorrect because the distribution of income (and wealth) lies within the purview of fiscal policy involving the government's decisions concerning spending and taxes. By contrast, monetary policy refers to central bank activities directed toward influencing the quantity of money and credit in an economy. Income distribution is not a policy domain of monetary policy.

C is incorrect because the overarching goal of monetary and fiscal policy is to create economic conditions characterized by positive, stable economic growth and low, stable inflation. Achieving this goal promotes stability (rather than cyclicality) in employment, consumption, and saving/investment outcomes.

Monetary and Fiscal Policy
LOS a
Section 1

46 An accounting document that records transactions in the order in which they occur is *best* described as a:

A trial balance.

B general ledger.

C general journal.

C is correct. The general journal records transactions in the order in which they occur (chronological order) and is thus sorted by date.

A is incorrect. The trial balance lists all account balances at a particular point in time.

B is incorrect. The general ledger is a document that shows all business transactions by account.

Financial Reporting Mechanics
LOS g
Section 6.1

47 Which of the following statements *best* describes the role of the International Organization of Securities Commissions (IOSCO)? The IOSCO

A is the oversight body to which the International Accounting Standards Board (IASB) reports.

B is responsible for regulating financial markets of member nations.

C assists in attaining the goal of cross-border cooperation in combating violations of securities laws.

C is correct. The IOSCO is not a regulator of financial markets. Its role is to assist in attaining the goal of uniform regulation and enforcement of international financial standards and in attaining the goal of cross-border cooperation in combating violations of securities and derivative laws.

A is incorrect. The IOSCO assists in attaining the goal of uniform regulation of international financial standards including IFRS, but the IASB does not report to it.

B is incorrect. The IOSCO is not a regulatory authority.

Financial Reporting Standards
LOS b
Section 3.2.1

48 The following information is available on a company for the current year.

Net income	$1,000,000
Average number of common shares outstanding	100,000
Details of convertible securities outstanding:	
Convertible preferred shares outstanding	2,000
Dividend/share	$10
Each preferred share is convertible into five shares of common stock	
Convertible bonds, $100 face value per bond	$80,000
8% coupon	
Each bond is convertible into 25 shares of common stock	
Corporate tax rate	40%

The company's diluted EPS is *closest* to:

A $7.72.

B $7.57.

C $7.69.

A is correct. Because both the preferred shares and the bonds are dilutive, they should both be converted to calculate the diluted EPS. Diluted EPS is the lowest possible value.

	Basic EPS	**Diluted EPS: Bond Converted**	**Diluted EPS: Preferred Converted**	**Diluted EPS: Both Converted**
Net income	$1,000,000	$1,000,000	$1,000,000	$1,000,000
Preferred dividends	–$20,000	–$20,000	0	0
After-tax cost of interest 8% × $80,000 × (1 – 0.40)		$3,840		$3,840
Numerator	$980,000	$983,840	$1,000,000	$1,003,840
Average common shares outstanding	100,000	100,000	100,000	100,000
Preferred converted			10,000	10,000
Bond converted		20,000		20,000
Denominator	100,000	120,000	110,000	130,000

	Basic EPS	Diluted EPS: Bond Converted	Diluted EPS: Preferred Converted	Diluted EPS: Both Converted
EPS	\$9.80	\$8.20	\$9.09	\$7.72

B is incorrect. Preferred dividends were included after preferred conversion (see table).

C is incorrect. After-tax interest on bonds was not added back after bond conversion: after-tax interest is (1 – 0.40) × 8% × \$80,000 = 3,840 (see table).

Understanding Income Statements
LOS h, i
Sections 6.2, 6.3

49 For which of the following inventory valuation methods is the gross profit margin *least likely* to be the same under both a perpetual inventory system and a periodic inventory system?

A LIFO

B Specific identification

C FIFO

A is correct. The periodic and perpetual systems result in the same inventory and cost of goods sold values (and thus gross profit margin) using both FIFO and specific identification valuation methods, but not always under LIFO.

B is incorrect. The periodic and perpetual systems result in the same inventory and cost of goods sold values (and thus gross profit margin) using both FIFO and specific identification valuation methods, but not always under LIFO.

C is incorrect. The periodic and perpetual systems result in the same inventory and cost of goods sold values (and thus gross profit margin) using both FIFO and specific identification valuation methods, but not always under LIFO.

Inventories
LOS c
Section 3.6

50 Under International Financial Reporting Standards (IFRS), which of the following financial statement elements *most* accurately represents inflows of economic resources to a company?

A Revenues

B Assets

C Owners' equity

A is correct. The financial statement elements under IFRS are assets, liabilities, owners' equity, revenue, and expenses. Revenues are inflows of economic resources. Assets are economic resources but not inflows. Equity is the residual claim on those economic resources.

B is incorrect. Assets are economic resources, but not inflows.
C is incorrect. Equity is a residual claim on economic resources.

Financial Reporting Mechanics
LOS b
Section 3

51 Under IFRS, which of the following balance sheet presentation formats is *most* acceptable? Classifying assets and liabilities:

A into operating, investing, and financing categories.

B in liquidity order.

C as monetary vs. non-monetary.

B is correct. A liquidity-based presentation can be used when it provides information that is reliable and more relevant. Entities that typically choose this format include banks.

A is incorrect. Cash flow statements are divided into these categories.

C is incorrect. The monetary/non-monetary categorization would be required in preparation for translating the balance sheet of a foreign subsidiary because different exchange rates apply. However, this approach is not an acceptable balance sheet reporting format.

Understanding Balance Sheets
LOS c
Section 2.2, 2.3

52 Previously, a manufacturer of high-quality industrial electrical generators only sold its units to customers, but it has just introduced a leasing program. The generators have expected useful lives of about 25 years, and the company anticipates that the leases will have a term of 20 years or more. The company reports under International Financial Reporting Standards. Which of the following statements about the first year of the new leasing program is *most* accurate?

A Regardless of how the company classifies the lease, its total cash flow and operating cash flow will be the same.

B If the lease is classified as an operating lease, the company's profits should be higher for a given leased asset than they would be under a finance lease.

C If the lease is classified as a finance lease, it will decrease the company's liquidity position compared with when the company was only selling its generators.

C is correct. Whether the company sells or leases the asset, inventory will be reduced. For sales, the company would report an accounts receivable classified as a current asset (assuming sales terms are not in question). If the leases qualify as finance leases, then the company will report a lease receivable, which is primarily long term. Therefore, compared with selling units outright, the company's current assets are lower under leasing and its liquidity position will decrease.

A is incorrect. Total cash flow over the lease term under the two different methods of lease accounting will be the same, but the total operating cash flows will not be: operating cash flows will be lower under the finance lease, but investing cash flows will be higher.

B is incorrect. Total profits over a lease term under the two different methods of lease accounting will be the same, but net income will be higher in the early years of the finance lease compared to the net of lease revenues less depreciation reported under an operating lease.

Long-Lived Assets
LOS p
Section 9.2.2
Non-Current (Long-Term) Liabilities
LOS g
Section 3.2.2

53 The *most* appropriate treatment for intangible assets with indefinite useful lives is to:

A expense.

B capitalize with no amortization.

C capitalize and amortize.

B is correct. Intangible assets assumed to have indefinite useful lives (i.e., no foreseeable limit to the period over which the asset is expected to generate net cash inflows for the company) are capitalized and not amortized.

A is incorrect. Acceptable amortization methods for intangible assets are the same as those for deprecation, but given that the intangible has an indefinite life, expensing it is not an option.

C is incorrect. Acceptable amortization methods for intangible assets are the same as those for deprecation, but given that the intangible has an indefinite life, the straight-line method is not an option.

Long-Lived Assets
LOS f
Section 3.2

54 The following data are available on a company:

Metric	
Working capital	$60 million
Non-current assets	$235 million
Equity	$170 million
Current ratio	1.75

The company's financial leverage is *closest* to:

A 1.7.

B 2.2.

C 1.2.

B is correct. First determine current assets, where CA = Current assets, CL = Current liabilities, WC = Working capital, and CR = Current ratio.

$$\begin{aligned} CR &= CA/CL = 1.75 \\ CL &= CA/1.75 \\ WC &= CA - CL \\ 60 &= CA - CA/1.75 \\ 60 &= 0.75/1.75 \times CA \\ CA &= 140 \end{aligned}$$

Then solve for total assets and determine financial leverage:

Metric	
Current assets	\$140 million
Non-current assets	+ \$235 million
Total assets	\$375 million
Equity	\$170 million
Financial leverage = Total assets/Equity	2.2

A is incorrect. Instead of calculating total assets, the sum of non-current assets and working capital was used:

Metric (all \$ amounts in millions)	
Working capital	60
Non-current assets	+ 235
Incorrect total assets	295
Equity	170
Financial leverage = Total assets/Equity	1.7

C is incorrect. Instead of calculating financial leverage, total liabilities to equity was calculated:

Metric (all \$ amounts in millions)	
Total assets per calculation above	375
Less: Equity	–170
Total liabilities	205
Incorrect Financial leverage calculation (Total liabilities/Equity)	1.2

Understanding Balance Sheets
LOS h
Section 7.2

55 Selected information from a company's comparative income statement and balance sheet is presented below:

Selected Income Statement Data for the Year Ended 31 August ($ thousands)	2013	2012
Sales revenue	100,000	95,000
Cost of goods sold	47,000	47,500
Depreciation expense	4,000	3,500
Net Income	11,122	4,556

Selected Balance Sheet Data as of 31 August ($ thousands)	2013	2012
Current Assets		
Cash and investments	21,122	25,000
Accounts receivable	25,000	13,500
Inventories	13,000	8,500
Total current assets	59,122	47,000
Current liabilities		
Accounts payable	15,000	15,000
Other current liabilities	7,000	9,000
Total current liabilities	22,000	24,000

The cash collected from customers in 2013 is *closest* to:

A $111,500.

B $96,100.

C $88,500.

C is correct.

$$\begin{aligned}\text{Cash collected from customers} &= \text{Revenues} - \text{Increase in accounts receivable}\\ &= \$100 - (25 - 13.5)\\ &= \$88.5 \text{ thousand}\end{aligned}$$

A is incorrect. It adds the increase in accounts receivable to sales: $100 + (25 – 13.5) = 111.5.

B is incorrect. It deducts the decrease in cash from revenues: 100 – (25 – 21.1) = 96.1.

Understanding Cash Flow Statements
LOS e, f
Section 3.2.1.1

56 The following information is available about a company for its current fiscal year:

Accounting profit (earnings before taxes)	$250,000
Taxable income	$215,000
Tax rate	30%
Income taxes paid in year	$61,200
Deferred tax liability, start of year	$82,400
Deferred tax liability, end of year	$90,650

The income tax expense reported on the current year's statement of earnings is *closest* to:

A $72,750.

B $69,450.

C $64,500.

A is correct. Income tax expense equals income tax payable (the tax rate multiplied by the taxable income) plus the increase in the deferred tax liabilities.

(0.30 × $215,000) + ($90,650 – $82,400) = $64,500 + $8,250 = $72,750

B is incorrect. It is the tax paid plus the increase in deferred taxes: 61,200 + 8,250 = 69,450.

C is incorrect it is just the tax rate × the taxable income: 30 × 215,000 = 64,500.

Income Taxes
LOS a, d
Section 2

57 On 1 January 2014, the market rate of interest on a company's bonds is 5%, and it issues a bond with the following characteristics:

Face value	€50 million
Coupon rate, paid annually	4%
Time to maturity	10 years (31 December 2023)
Issue price (per €100)	€92.28

If the company uses International Financial Reporting Standards (IFRS), its interest expense (in millions) in 2014 is *closest* to:

A €2.307.

B €2.386.

C €1.846.

A is correct. IFRS requires the effective interest method for the amortization of bond discounts/premiums. The bond is issued for 0.9228 × €50 million = €46.140.

Interest expense = Liability value × Market rate at issuance = 0.05 × €46.140 = €2.307

C is incorrect. It uses the coupon rate of 4% × 46.140 = 1.8456.

B is incorrect. It uses the straight-line method of bond amortization, which is allowed under US GAAP but not IFRS.

$$50 \times 4\% - [(100 - 92.28) \times 50]/10 \text{ years} = 2 + 386{,}000 = 2.386$$

Non-Current (Long-Term) Liabilities
LOS b
Section 2.2

58 A company that prepares its financial statements according to US GAAP leased a piece of equipment on 1 January 2013. Information relevant to the transaction is as follows:

- Five annual lease payments of $25,000, with the first payment due 1 January 2013
- Interest rate on similar company debt is currently 8%
- The fair value of the equipment is $115,000
- Useful life of the equipment is seven years
- The company depreciates other equipment in the same asset class on a straight-line basis

The total expense related to the lease on the company's 2013 income statement will be *closest* to:

A $25,000.

B $28,185.

C $22,024.

B is correct. The lease would qualify as a finance (capital) lease under US GAAP because the present value (PV) of the lease payments is more than 90% of the fair value.

Using a financial calculator for an annuity due at the beginning of the period:

PV of lease payments: PMT = $25,000, i = 8%, N = 5, Mode = Begin, Compute PV.

PV = $107,803, 90% of the fair value: 0.90 × $115,000 = $103,500

Therefore, the lease is greater than 90% and would be capitalized at $107,803.

Present value of the lease (asset value capitalized and initial liability)		$107,803
Payment 1 January 2013		–25,000
Liability value 1 January 2013		$82,803
Interest expense in 2013	0.08 × $82,803	$6,624.25
Amortization expense for the year using the lease term as the useful life (no indication that the lease will be renewed beyond the initial term)	$107,803/5	$21,560.63
Total expense in 2013		$28,184.88

A is incorrect. It assumes it is an operating lease and simply deducts the lease payment. If you do not calculate the PV as an annuity due then it does not trip the criteria and you would incorrectly classify it as operating. N = 5, i = 8, PMT = 25,000, PV = 99,817 vs. 90% of 115,000 = 103,500.

C is incorrect. It correctly classifies it as a finance lease but amortizes it over 7 years: 107,803/7 = 15,400; 15,400 + 6,624 = 22,024.

Non-Current (Long-Term) Liabilities
LOS g, h
Section 3.2.1

59 An equity analyst is forecasting next year's net profit margin of a heavy equipment manufacturing firm by using the average net profit margin over the past three years. In making his profit projection, he identifies the following three items:

1. The company reported losses from discontinued operations in each of the past three years.
2. The most recent year's tax rate was only half of the prior two years' rate as a result of a fiscal stimulus.
3. The company reported gains on the sale of investments in each of the past three years.

Which of the following statements about the preparation of the forecast is *most* accurate? The analyst would:

A use the most recent tax rate because it is the best predictor of future tax rates.

B exclude the gains on the sale from investments because the company is a manufacturing firm.

C include the losses from discontinued operations because they appear to be an ongoing feature for this company.

B is correct. The company is a heavy equipment manufacturer. Because gains on investments are not a core part of the company's business, they should not be viewed as an ongoing source of earnings. Discontinued operations are considered to be non-recurring items (even though they have occurred in the past three years); they are normally treated as random and unsustainable and should not be included in a short-term forecast. The change in the current tax rate is best viewed as temporary in the absence of additional information and should not be the basis of the calculation of the average tax rate.

A is incorrect. The long(er) run tax rate should be used; the change in the current tax rate is best viewed as temporary in the absence of additional information and should not be the basis of the calculation of the average tax rate.

C is incorrect. The discontinued items should be considered random and unsustainable for a short-term forecast.

Understanding Income Statements
LOS f
Sections 5.1, 5.5
Financial Statement Analysis: Applications
LOS b
Section 3

60 A company has consistently and significantly increased its cash balance over the past three years. The *least likely* explanation for the increase in cash is a:

A forthcoming issue of new equity.

B potential acquisition.

C planned increase in the dividend.

A is correct. A new equity issue will provide additional cash, so it does not explain an increase in cash holdings prior to the issue.

B is incorrect. A company might increase its cash to build a "war chest" in order to make an acquisition.

C is incorrect. A company might increase its cash to pay a larger dividend.

Financial Statement Analysis: Applications
LOS a
Section 2

61 The financial statement that would be *most* useful to an analyst in understanding the changes that have occurred in a company's retained earnings over a year is the statement of:

A comprehensive income.

B financial position.

C changes in equity.

C is correct. The statement of changes in equity reports the changes in the components of shareholders' equity over the year, which would include the retained earnings account.

A is incorrect. The net income determined in the calculation of comprehensive income is a component of the change in retained earnings, but there are other changes that may also have occurred (the payment of dividends, for example) that are not included on the statement of comprehensive income.

B is incorrect. Although the year-end balances of retained earnings may be reported on the statement of financial position (depending on if the company breaks out the components of shareholders' equity), it would not detail the changes over the year.

Financial Statement Analysis: An Introduction
LOS b
Section 3.1.3

62 Under US GAAP, interest paid is *most likely* included in which of the following cash flow activities?

A Operating only

B Financing only

C Either operating or financing

A is correct. Interest paid must be categorized as an operating cash flow activity under US GAAP, although it can be categorized as either an operating or financing cash flow activity under IFRS.

B is incorrect. Interest paid cannot be categorized as a financing cash flow under US GAAP but can be under IFRS.

C is incorrect. Interest paid cannot be categorized as a financing cash flow under US GAAP but can be under IFRS.

Understanding Cash Flow Statements
LOS a, c
Section 3.2.1.5

63 A firm reported the following financial statement items:

	€ millions
Net income	2,100
Non-cash charges	400
Interest expense	300
Capital expenditures	210
Working capital expenditures	0
Net borrowing	1,600
Tax rate	40%

The free cash flow to the firm (FCFF) is *closest* to:

A €2,110.

B €2,470.

C €2,590.

B is correct.

$$FCFF = NI + NCC + Int(1 - t) - FCInv - WCInv$$

	Cash Flow Item		**Amount (€ millions)**
NI	Net income		2,100
NCC	Plus non-cash charges		400
$Int(1 - t)$	Plus interest expense (1 – tax rate)	300 (1 – 0.40)	180
FCInv	Less capital expenditures		(210)
WCInv	Less working capital expenditures		0
FCFF	Free cash flow to the firm		**€2,470**

A is incorrect. It incorrectly subtracted interest expense.

Cash Flow Item		**Amount (€)**
Net income	+	€2,100
Non-cash charges	+	400
Capital expenditure	–	210
Interest Expense: Int × $(1 - t)$	–	180
FCFF		€2,110

C is incorrect. It ignores the tax effect on interest expense.

Cash Flow Item	Amount (€)
Net income	2,100
Plus Non-cash charges	400
Plus Interest expense	300
Less Capital expenditure	(210)
Less working capital expenditures	0
FCFF	**€2,590**

Understanding Cash Flow Statements
LOS i
Section 4.3

64 The following information is from a company's accounting records:

	€ millions
Revenues for the year	12,500
Total expenses for the year	10,000
Gains from available-for-sale securities	1,475
Loss on foreign currency translation adjustments on a foreign subsidiary	325
Dividends paid	500

The company's total comprehensive income (in € millions) is *closest* to:

A 1,150.

B 3,150.

C 3,650.

C is correct.

Total comprehensive income = Net income + Other comprehensive income

Net Income = Revenues – Expenses.

Other comprehensive income includes gains or losses on available-for-sale (AFS) securities and translation adjustments on foreign subsidiaries.

(Revenues – Expenses) + Gain on AFS securities – Loss on FX translation

(12,500 – 10,000) + 1,475 – 325 = 3,650.

A is incorrect because it is just the other comprehensive income (1,475 – 325) and not the total comprehensive income.

B is incorrect because it also deducts dividends, which are deducted when calculating the change in shareholders' equity but not total comprehensive income.

Understanding Income Statements
LOS l, m
Section 8

65 Under US GAAP, which of the following is *least likely* a disclosure concerning inventory?

A The amount of inventories recognized as an expense during the period

B The carrying amounts of inventories carried at fair value less costs to sell

C The amount of the reversal of any write-down of inventories

C is correct. US GAAP does not permit the reversal of prior year write-downs; therefore, there are no disclosures related to reversals.

A is incorrect. Under both US GAAP and IFRS, disclosure is required about the amount of inventories expensed during the period (cost of goods sold).

B is incorrect. Both US GAAP and IFRS require disclosures of any inventories carried at fair value less costs to sell (NRV).

Inventories
LOS i
Section 7.1

66 Which of the following accounting actions would increase stockholders' equity in the current period?

A Using LIFO rather than FIFO accounting for inventory in an inflationary environment.

B Capitalizing, rather than expensing, a payment.

C Increasing the allowance for uncollectible accounts receivable.

B is correct. The capitalization of payments is an example of how choices affect both the balance sheet and income statement. Capitalizing a payment changes the benefit from only the current period—making it an expense—to a benefit in future periods as an asset. The creation of an asset results in a comparable increase in stockholder's equity.

A is incorrect because adopting last-in,-first-out accounting for inventory in an inflationary environment assumes newer and costlier purchases of inventory are sold to customers with the result that the remaining inventory reflects the older costs. This will contribute to lower net income from sales and a decrease in stockholders' equity

C is incorrect because increasing the allowance for uncollectible accounts receivable will increase the uncollectible accounts expense reported for the period. This will contribute to lower net income and a decrease in stockholders' equity.

Financial Reporting Quality
LOS h
Section 4.2.1

67 Conservative, rather than aggressive, accounting is most likely associated with:

A increased sustainability of earnings.

B higher current reported performance.

C recognition of losses once certain.

A is correct. Conservative accounting choices decrease a company's reported performance and results in the current period and may increase its reported performance and financial position in later periods. Therefore, it typically avoids a sustainability issue.

B is incorrect because higher current reported performance is a result associated with aggressive accounting choices, not conservative ones.

C is incorrect because in general, conservatism means that losses are recognized when probable; waiting to recognize losses until they are certain would be an aggressive approach rather than a conservative one.

Financial Reporting Quality
LOS c
Section 2.5.1

68 Using the data below, an analyst is in the process of comparing two companies (A and B) that are in the same industry.

Company A (in $ millions, except per share data)	
Net income	$7,098
Weighted average common shares outstanding	4,366
Common share dividends	$1,700
Stock price per share at year-end	$41.00

Company B	
Basic EPS	$4.35
Earnings multiple	21.17×
Dividend payout ratio	25.9%

Compared with Company B, Company A *most likely* has a higher:

A price-to-earnings ratio.
B dividend payout ratio.
C earnings per share.

A is correct. Company A has a higher price-to-earnings ratio (P/E). Its P/E of 25.15× is calculated as follows:

	Calculation	Company A	Comment
EPS	Net income/Weighted average number of common shares outstanding = 7,098/4,366 =	\$1.63	Lower than B (\$4.35)
P/E	Market price/EPS = 41.00/1.63 =	25.15×	**Higher than B (21.17×)**
Dividend payout	Common share dividends/Net income = 1,700/7,098 =	24.0%	Lower than B (25.9%)

B is incorrect because Company A has a lower dividend payout than Company B: 24.0% vs. 25.9%.

C is incorrect because Company A has a lower EPS than Company B: 1.63 vs. 4.35.

Financial Analysis Techniques
LOS e
Section 5.1
Understanding Income Statements
LOS h
Section 6.2

69 The following information is available for a company:

	2016 (in € millions)
EBIT (earnings before interest and taxes)	1,015.0
Interest expense	73.4
Tax expense	201.4
Total assets	5,305.0
Average total assets	5,421.0
Total debt	1,048.0

	2015
Interest coverage	15.3×
Debt to total assets	18.2%
Operating return on assets (ROA)	17.3%

Compared with 2015, which of the following ratios *most likely* indicates an improvement in the creditworthiness of the company? The change in the company's:

A operating ROA.

B debt-to-total assets.

C interest coverage.

A is correct. When calculated for 2016, interest coverage decreased and debt to total assets and return on assets both increased. In general, a decrease in interest coverage and an increase in debt to total assets would reduce creditworthiness, but an increase in return on assets would improve creditworthiness. Calculations are as follows:

	Calculation	2016	Comment
Debt to total assets	Total debt/ Total assets = €1,048/€5,305 =	19.8%	Higher than 2015 (18.2%), which would reduce creditworthiness
Interest coverage	EBIT/Interest payments = €1,015/€73.4 =	13.8×	Lower than 2015 (15.3×), which would reduce creditworthiness
Operating ROA	EBIT/Average total assets = €1,015/€5,421 =	18.7%	**Higher than 2015 (17.3%), which would improve creditworthiness**

B is incorrect because an increase in debt-to-total assets would likely reduce creditworthiness.

C is incorrect because a decrease in the interest coverage ratio would likely reduce creditworthiness.

Financial Analysis Techniques
LOS e
Section 6

70 A 20-year $1,000 fixed-rate non-callable bond with 8% annual coupons currently sells for $1,105.94. Assuming a 30% marginal tax rate and an additional risk premium for equity relative to debt of 5%, the cost of equity using the bond-yield-plus-risk-premium approach is *closest* to:

A 9.9%

B 12.0%

C 13.0%

B is correct. First, determine the yield to maturity, which is the discount rate that sets the bond price to $1,105.94 and is equal to 7%. This calculation can be done with a financial calculator:

FV = –$1,000, PV = $1,105.94, N = 20, PMT = –$80, solve for i, which will equal 7%.

The bond-yield-plus-risk-premium approach is calculated by adding a risk premium to the cost of debt (i.e., the yield to maturity for the debt), making the cost of equity 12.00% (= 7% +5%).

A is incorrect because it uses the after-tax cost of debt: 9.90% = 7% × (1 – 30%) + 5%.

C is incorrect because it uses the coupon rate instead of the yield-to-maturity: 13.00% = 8% + 5%.

Cost of Capital
LOS f, h
Sections 3.1.1, 3.3.3

71 A company's data are provided in the following table:

Cost of debt	10%
Cost of equity	16%
Debt-to-equity ratio (D/E)	50%
Tax rate	30%

The weighted average cost of capital (WACC) is *closest* to:

A 14.0%.

B 11.5%.

C 13.0%.

C is correct. Convert the D/E to determine the weights of debt and equity as follows:

$$w_d = \frac{D/E}{1 + D/E} = \frac{50\%}{1 + 50\%} = 33.3\%$$

$$w_e = 1 - w_d = 66.7\%$$

$$\begin{aligned}\text{WACC} &= w_d r_d (1 - t) + w_p r_p + w_e r_e \\ &= 33.3\% \times 10\% \times (1 - 30\%) + 66.7\% \times 16\% = 13.0\%\end{aligned}$$

A is incorrect because the debt is not tax adjusted when determining the WACC.

$$\text{WACC} = 33.3\% \times 10\% + 66.7\% \times 16\% = 14.0\%$$

B is incorrect because the D/E is used as the weight for debt and equity.

$$\text{WACC} = 50\% \times 10\% \times (1 - 30\%) + 50\% \times 16\% = 11.5\%$$

Cost of Capital
LOS a, b
Sections 2, 2.1, 2.2

72 A company is considering a switch from an all-equity capital structure to a structure with equal amounts of equity and debt without increasing assets. This change will reduce the net income by 30%. If the current return on equity (ROE) is 10%, the ROE under the proposed capital structure will be *closest* to:

A 6%.

B 20%.

C 14%.

C is correct.

All Equity	**Half Equity and Debt**
Net income	Net income × (1 – 30%)
Equity = 100% × Assets	Equity = 50% × Assets
ROE: = Net income/equity = Net income/(100% × Assets) = Net income/Assets = 10%	ROE: = [Net income × (1 – 30%)]/Equity = [Net income × (1 - 30%)]/(50% × Assets) = (Net income/Assets) × [(1 – 30%)/50%] = 10% × 1.4 = 14%

Alternatively, looking at the effects of the changes in sequence:

When equity decreases by half, ROE would become 10%/0.50 = 20%.

When net income decreases by 30%, the adjusted ROE would become = 20% × (1 – 0.30) = 14%.

A is incorrect because it uses (Net Income × 30%) instead of (Net Income × [1 – 30%]) in the solution. (Net Income/Assets) × (30%/50%) = 10% × 0.6 = 6%.

B is incorrect because it does not adjust net income in the solution.

$$(\text{Net Income/Assets})/50\% = 20\%$$

Measures of Leverage
LOS c
Section 3.4

73 The following information is available for a firm in a developing country:

Risk-free rate	2.0%
Firm's equity beta	1.5
Equity risk premium in a developed country	3.0%
Developing country risk premium	4.0%
Sovereign yield spread	2.5%

The firm's cost of equity using the CAPM approach is *closest* to:

A 10.5%.
B 12.5%.
C 10.3%.

B is correct. Cost of equity = Risk-free rate + Equity beta × (Equity risk premium + Country risk premium) = 0.02 + 1.5 × (0.03 + 0.04) = 12.5%.

C is incorrect. Uses sovereign yield spread instead of equity premium.

$$0.02 + 1.5 \times (0.03 + 0.025) = 10.25\%.$$

A is incorrect. The developing country risk premium is added:

$$0.02 + 1.5(0.03) + 0.04 = 10.5$$

Or

$$(0.03 + 0.04) \times 1.5, \text{ ignoring the risk free rate}$$

Cost of Capital
LOS j, h
Section 4.2 and 3.3

74 A company extends its trade credit terms by four days to all its credit customers. These credit customers are *most likely* to experience a four-day:

A decrease in their net operating cycle.
B increase in their operating cycle.
C decrease in their operating cycle.

A is correct. The company's customers are receiving a four-day increase in their number of days of payables, which will reduce the company's cash conversion cycle (net operating cycle) by four days.

B is incorrect because a four-day increase in the number of days of payables will not affect the operating cycle (DHO + DSO) but will reduce the net operating cycle by four days

C is incorrect because a four-day increase in the number of days of payables will not affect the operating cycle (DHO + DSO) but will reduce the net operating cycle by four days.

Financial Analysis Techniques
Section 4.2-4.3
LOS b
Working Capital Management
LOS c
Section 2.2

75 Which of the following statements describes the *most* appropriate treatment of cash flows in capital budgeting?

A Interest costs are included in the project's cash flows to reflect financing costs.

B A project is evaluated using its incremental cash flows on an after-tax basis.

C Sunk costs and externalities should not be included in the cash flow estimates.

B is correct. All of the incremental cash flows arising from a project should be analyzed on an after-tax basis.

C is incorrect. Only sunk costs should be ignored in a project's cash flow estimation, but not any externalities. Sunk costs cannot be recovered once they have been incurred. Externalities (both positive and negative ones) are the effects of an investment decision on other things beside the investment itself; they should therefore be included in the cash flow estimation.

A is incorrect. Financing costs like interest costs are excluded from calculations of operating cash flows. The financing costs are reflected in the required rate of return for an investment project. If financing costs are included, we would be double-counting these costs.

Capital Budgeting
LOS b
Section 3

76 Based on a need to borrow $2 million for one month, which of the following alternatives has the *least* expensive effective annual cost?

A A credit line at 6.0% annually with a $4,000 annual commitment fee

B A banker's acceptance with an all-inclusive annual rate of 6.1%

C Commercial paper at 5.9% annually with a dealer's annual commission of $3,000 and a backup line annual cost of $4,000

B is correct.

Method	Formula	Calculation
Banker's Acceptance (BA)	$\frac{\text{Interest}}{\text{Net proceeds}} \times 12$	Interest: 0.061/12 × \$2,000,000 = 10,167 Net proceeds: 2,000,000 – 10,167 = 1,989,833 BA cost (10,167 × 12)/1,989,833 = **0.0613**
Line of credit (LOC)	$\frac{\text{Interest + Commitment fee}}{\text{Usable loan amount}} \times 12$	Interest: 0.06/12 × \$2,000,000 = 10,000 Commitment fee: \$4,000/12 = 333 Usable loan: \$2,000,000 LOC cost (10,333 × 12)/2,000,000 = **0.0620**
Commercial Paper (CP)	$\frac{\text{Interest + Dealer's commissions + Backup costs}}{\text{Net proceeds}} \times 12$	Interest: 0.059/12 × \$2,000,000 = 9,833 Dealer commission: \$3,000/12 = 250 Backup costs: \$4,000/12 = 333 Total costs: 9,833 + 250 + 333 = 10,416 Net proceeds: 2,000,000 – 9,833 = 1,990,167 CP cost: (10,416 × 12)/1,990,167 = **0.0628**

Banker's acceptance has the lowest annual effective cost of 0.0613.
A is incorrect because the line of credit effective annual cost is 6.20%.
C is incorrect because the commercial paper effective annual cost is 6.28%.

Working Capital Management
LOS g
Section 8.4

77 Which of the following *best* allows a board of directors to act in the interest of the company and shareholders?

A Independent board members are selected from outside the industry.

B Internal directors provide monitoring of the firm's management.

C The board has the authority to select and terminate senior management.

C is correct. The board has a duty to make decisions in the best interest of the company and shareholders. It reviews management's performance in executing the strategy set by the board. In order to ensure strong execution of the strategy, the board must have the authority to select and terminate senior management.

A is incorrect. At least some independent members should have industry expertise.
B is incorrect. External directors should provide monitoring within the firm.

Corporate Governance and ESG: An Introduction
LOS f
Section 5.2

78 A project has the following cash flows (\$) where the cash inflows are earned evenly throughout the year:

Year 0	Year 1	Year 2	Year 3	Year 4
–1,525	215	345	475	1,215

Assuming a discount rate of 11% annually, the discounted payback period (in years) is *closest* to:

A 3.4.

B 3.9.

C 4.0.

B is correct. The discounted cash flows (CF) and their cumulative sum are:

Year	Discounted CF	Cumulative Discounted CF	Cumulative Cash Inflows
0	–1,525.00	–1,525.00	
1	$193.69 = 215/(1.11)^1$	–1,331.31	193.69
2	$280.01 = 345/(1.11)^2$	–1,051.30	473.70
3	$347.32 = 475/(1.11)^3$	–703.98	821.02
4	$800.36 = 1{,}215/(1.11)^4$	96.38	1,621.38

After three years, $821.02 of the $1,525 investment is recovered leaving, $703.98 left to recover in the fourth year. Proportionately, only 0.88 (= $703.98/$800.36) of the cash flow in the fourth year is necessary to recover all of the investment, which makes the discounted payback equal to 3.9 years (rounded up from 3.88).

A is incorrect. This is the calculation with the payback period, which is not discounted.

C is incorrect. Based on the above calculation, the discounted payback to 3.88 years, the nearest round off value should be 3.9.

Capital Budgeting
LOS d
Section 4.4

79 The execution step of the portfolio management process includes:

A preparing the investment policy statement.

B finalizing the asset allocation.

C monitoring the portfolio performance.

B is correct. Asset allocation occurs in the execution step.

A is incorrect. Preparation of the investment policy statement occurs in the planning step.

C is incorrect. Portfolio monitoring occurs in the feedback step.

Portfolio Management: An Overview
LOS d
Section 4

80 An investor with $10,000 decides to borrow an additional $5,000 at the risk-free rate and invest all the available funds in the market portfolio. This investor's portfolio beta is *closest* to:

A 0.5.

B 1.0.

C 1.5.

C is correct. The weight in the market portfolio is 15,000/10,000 = 1.5 and the weight in the risk-free asset is –5,000/10,000 = –0.5. Because the beta of the risk-free asset is 0 and the market portfolio's beta is 1, the portfolio's beta is $\beta_p = 0(-0.5) + 1(1.5) = 1.5$.

A is incorrect because it is computed using weights of 0.5 for the risk-free asset and the market portfolio, that is: 0(0.5) + 1(0.5) = 0.5.

B is incorrect because it is the market portfolio's beta.

Portfolio Risk and Return: Part II
LOS b, e
Sections 2.2 and 4.2

81 The variance of returns of a security and the market portfolio are 0.25 and 0.09, respectively. If the covariance of security returns and market returns is 0.06, the security's beta is *closest* to:

A 0.24.

B 0.67.

C 0.40.

B is correct. The security's beta is:

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\sigma_m^2} = \frac{0.06}{0.09} = 0.67$$

A is incorrect because it is computed as (0.06/0.25) = 0.24.

C is incorrect because it is computed as the correlation between the returns on the security and the market portfolio as [0.06/(0.5 × 0.3)] = 0.4.

Portfolio Risk and Return: Part II
LOS e
Section 3.2

82 Risk management is *most likely* the process by which an organization:

A minimizes its exposure to potential losses.

B adjusts its risk to a predetermined level.

C maximizes its risk-adjusted return.

B is correct. An organization with a strong competitive position can recover from losses more easily than one with a weaker competitive position. Therefore, an organization's risk tolerance should reflect its competitive position. An organization's size does not define the risk sources it faces or the relative losses it can absorb, so it should not be reflected in its risk tolerance. Neither the risk sources affecting an organization nor the size of the losses an organization can absorb are a function of its perception of market stability.

A is incorrect because a minimum risk position may not be optimal for the organization.

C is incorrect because a high risk-adjusted return may require a level of risk taking that is beyond the organization's capacity to absorb.

Risk Management: An Introduction
LOS a
Section 3.2

83 A good risk governance process would *most likely*:

A provide guidance on the size of the largest acceptable loss for the organization.

B provide different risk targets for each unit within the organization.

C be a bottom-up process that reflects the current risk exposures of all parts of the organization.

A is correct. A quality risk governance process takes a top-down approach and is charged with risk oversight for the entire organization. It should operate on an enterprise-wise basis rather than viewing each unit in isolation. It will determine the organization's risk tolerance and provide a sense of the maximum loss the organization can absorb.

B is incorrect because a good risk governance process looks at the enterprise as a whole rather than viewing individual units in isolation.

C is incorrect because a risk governance process is a top-down process.

Risk Management: An Introduction
LOS c
Section 3

84 Which of the following is *most likely* associated with an investor's ability to take risk rather than the investor's willingness to take risk?

A The investor has a long investment time horizon.

B The investor believes earning excess returns on stocks is a matter of luck.

C Safety of principal is very important to the investor.

A is correct. Investment time horizon is an objective factor that measures the investor's ability to take risk.

B is incorrect because luck is a subjective factor that measures willingness to take risk.

C is incorrect because safety of principal is a subjective factor that measures willingness to take risk.

Basics of Portfolio Planning and Construction
LOS d
Section 2.2.1

85 Which of the following portfolios of risky assets is *most likely* the global minimum-variance portfolio?

Portfolio	Expected Return	Standard Deviation
A	5%	20%
B	8%	33%
C	3%	20%

A Portfolio C

B Portfolio A

C Portfolio B

B is correct. The global minimum-variance portfolio is the left-most point on the minimum-variance frontier among all portfolios of risky assets. Portfolios A and C have the same standard deviation, but Portfolio A dominates Portfolio C because of a higher return.

A is incorrect because it has a lower expected return while having the same variance as portfolio A. It is unlikely to be located on the minimum-variance frontier.

C is incorrect because it has a higher variance than portfolio A.

Portfolio Risk and Return: Part I
LOS f
Section 5.2

86 An analyst observes that stock markets usually demonstrate return distributions concentrated to the right with a higher frequency of negative deviation from the mean. This feature is *most likely* known as:

A positive skewness.

B negative skewness.

C kurtosis.

B is correct. The negatively skewed investment characteristic is usually related to the stock returns whose distribution is concentrated to the right.

A is incorrect positive skew has more large positive deviation from the mean.

C is incorrect because kurtosis is related to the fat-tail feature.

Portfolio Risk and Return: Part I
LOS b
Section 2.5

87 Returns from a depository receipt are *least likely* affected by which of the following factors?

A Exchange rate movements

B Number of depository receipts

C Analysts' recommendations

B is correct. The price of each depository receipt (and, in turn, returns) will be affected by factors that affect the price of the underlying shares—such as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements. The number of depository receipts issued affects their price but does not affect the returns.

A is incorrect. The price of each depository receipt (and, in turn, returns) will be affected by such factors as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements.

C is incorrect. The price of each depository receipt (and, in turn, returns) will be affected by such factors as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements.

Overview of Equity Securities
LOS d, e
Section 5.2

88 Which of the following is *most likely* one of the main functions of the financial system?

A Determining an equilibrium interest rate

B Ensuring that markets are informationally efficient

C Ensuring that all investment projects receive sufficient funding

A is correct. One of the main functions of the financial system is to determine the equilibrium interest rate, which is the only interest rate that would exist if all securities were equally risky, had equal terms, and were equally liquid.

B is incorrect. Informational market efficiency is not a key function of the financial system, rather that of regulatory framework

C is incorrect. The financial system provides sufficient funding only to the most productive projects. An important function of the financial system is to direct resources away from wealth-diminishing projects.

Market Organization and Structure
LOS a
Section 2.2

89 A portfolio manager analyzes a market and discovers that it is not possible to achieve consistent and superior risk-adjusted returns, net of all expenses. This market is *most likely* characterized by:

A persistent anomalies.

B informational efficiency.

C restrictions on short selling.

B is correct. In an informationally efficient market, consistent and superior risk-adjusted returns (net of all expenses) are not achievable.

A is incorrect. In an informationally efficient market, consistent and superior risk-adjusted returns (net of all expenses) are not achievable. In such a situation, persistent anomalies are unlikely.

C is incorrect. Some market experts argue that restrictions on short selling limit arbitrage trading, which impedes market efficiency.

Market Efficiency
LOS a
Section 2.1

90 The financial systems that are operationally efficient are *most likely* characterized by:

A security prices that reflect fundamental values.

B liquid markets with low commissions and order price impacts.

C the use of resources where they are most valuable.

B is correct. Operationally efficient markets are liquid markets in which the costs of arranging trades, commissions, bid–ask spreads, and order price impacts are low.

A is incorrect. Informationally efficient markets, not the operationally efficient markets, are characterized by prices that reflect fundamental values so that prices vary primarily in response to changes in fundamental values and not to demands for liquidity made by uninformed traders.

C is incorrect. Allocationally efficient markets, not the operationally efficient markets, are characterized by using resources where they are most valuable.

Market Organization and Structure
LOS k
Section 9

91 According to the industry life-cycle model, an industry in the shakeout stage is *best* characterized as experiencing:

A little or no growth and industry consolidation.

B slowing growth and intense competition.

C relatively high barriers to entry and periodic price wars.

B is correct. The shakeout stage is usually characterized by slowing growth, intense competition, and declining profitability. During the shakeout stage, demand approaches market saturation levels because few new customers are left to enter the market. Competition is intense as growth becomes increasingly dependent on market share gains.

A is incorrect. Little or no growth and industry consolidation are characteristics of an industry in mature phase.

C is incorrect. Relatively high barriers to entry and periodic price wars are characteristic of mature phase.

Introduction to Industry and Company Analysis
LOS h
Section 5.1.5

92 In the semi-strong-form of market efficiency, fundamental analysis *most likely* requires the analyst to:

A extrapolate historical data to estimate future values and make investment decisions.

B use trading rules for detecting the price movements that lead to new equilibrium prices.

C do a superior job of estimating the relevant variables and predicting earnings surprises.

C is correct. Fundamental analysis facilitates a semi-strong-form efficient market by disseminating value-relevant information. Fundamental analysis can be profitable in terms of generating abnormal returns if the analyst creates a comparative advantage with respect to this information. Such an advantage can be achieved by doing a superior job of estimating the relevant variables and predicting earnings surprises.

A is incorrect. Simply extrapolating historical data may not produce superior returns in an efficient market.

B is incorrect. The use of technical rules for detecting the price movements and gradual price adjustments comprises technical analysis, not fundamental analysis.

Market Efficiency
LOS d
Section 3.4.1

93 A price-weighted index series is composed of the following three stocks:

Stock	Price before Split End of Day 1	Price after Split End of Day 2
X	\$10	\$12
Y	\$20	\$19
Z	\$60	\$22

If stock Z completes a three-for-one split at the end of Day 1, the value of the index after the split (at the end of Day 2) is *closest* to:

A 31.7.

B 32.3.

C 29.9.

A is correct. The value of the price-weighted index is determined by dividing the sum of the security values by the divisor, which is typically set at inception to equal the initial number of securities in the index. In the case of a stock split, the index provider must adjust the value of the divisor by dividing the sum of the constituent prices after the split by the value of the index before the split. This adjustment results in a new divisor that keeps the index value at the same level as before the split. The new divisor will then be used to calculate the index value after the split.

$$\text{Index before the split} = \frac{10 + 20 + 60}{3}$$

$$= 30$$

$$\text{New divisor, } X\text{: } 30 = \frac{10 + 20 + 20}{X}$$

$$X = 1.67$$

$$\text{Index after the split} = \frac{12 + 19 + 22}{1.67}$$

$$= 31.7$$

B is incorrect. It calculates the changes in prices and uses it to derive the new index.
C is incorrect. It uses price after split to calculate the new divisor.

Security Market Indexes
LOS b, e
Section 3.2.1

94 A firm reports negative earnings for the year just ended. The price multiple of the firm's stock that is *least likely* to be meaningful is:

A trailing price to earnings.

B price to cash flow.

C leading price to earnings.

A is correct. Negative earnings in the last year result in a negative ratio of trailing price to earnings and are not meaningful. Practitioners may use the ratio of (1) current price to cash flow or (2) leading price to earnings by replacing last year's loss with forecasted earnings.

B is incorrect. Alternative to negative trailing price-to-earnings ratio, practitioners may use price-to-cash-flow ratio because it is possible cash flow would be positive in spite of a small loss.

C is incorrect. Alternative to negative trailing price-to-earnings ratio, practitioners may use leading price-to-earnings ratio by replacing last year's loss with forecasted earnings which may be positive.

Equity Valuation: Concepts and Basic Tools
LOS j
Section 5

95 Which of the following types of companies is *most* sensitive to economic conditions?

A Cyclical

B Growth

C Defensive

A is correct. The performance of companies with cyclical demand for their products is highly variable and depends on economic conditions.

B is incorrect. Growth firms have steady increases in demand and are less dependent on economic conditions.

C is incorrect. Defensive demand is stable and does not fluctuate with the economic cycle, except in severe recessions.

Introduction to Industry and Company Analysis
LOS c
Section 5.1.7

96 Which of the following is the *most* appropriate reason for using a free cash flow to equity (FCFE) model to value equity of a company?

A FCFE models provide more accurate valuations than the dividend discount model.

B A firm's borrowing activities could influence dividend decisions, but they would not affect FCFE.

C FCFE is a measure of the firm's dividend paying capacity.

C is correct. FCFE is a measure of the firm's dividend-paying capacity.

A is incorrect. The statement that FCFE models provide more accurate valuations than the dividend discount models is not necessarily true. The appropriateness and the effectiveness of a model depend on the firm characteristics and the analyst's ability in making predictions.

B is incorrect. A firm's borrowing activities do impact FCFE, as in the expression:

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}.$$

Equity Valuation: Concepts and Basic Tools
LOS e
Section 4

97 An index provider launches a new index that will include value stocks in a specific country. This index will *most likely* be a:

A large-capitalization index.

B style index.

C fundamentally weighted index.

B is correct. Style indexes represent a group of securities classified according to market capitalization, value, and growth or a combination of these characteristics. Therefore, the new index will most likely be a style index with a value classification.

A is incorrect. Large-capitalization indexes represent and track the largest securities in terms of market capitalization. They do not represent the category of value stocks.

C is incorrect. Fundamentally weighted indexes use measures such as book value, cash flow, revenues, earnings, dividends, and number of employees to weight the constituent securities. Securities of different styles and sectors could be included in the same fundamentally weighted index. Therefore, fundamentally weighted indexes do not represent the category of value stocks.

Security Market Indexes
LOS h
Section 5.4

98 Given the following information for a company:

- Market value per share $250
- Current dividend per share $5
- Dividend growth rate 4%
- Required rate of return 6%

and using the Gordon growth model to estimate the intrinsic value, a share of the company is *best* described as being:

A fairly valued.

B overvalued.

C undervalued.

C is correct. The intrinsic value of an equity security based on the Gordon growth model is estimated as follows:

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

where:

V_0 = the estimate of the intrinsic value
D_0 = the current dividend ($5) and D_1 is the next year's dividend
g = the dividend growth rate (4%)
r = the required rate of return (6%)

The estimate of the intrinsic value is:

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{\$5(1+0.04)}{(0.06-0.04)} = \$260 \text{ per share}$$

Given that the market value ($250) is lower than the estimate of the intrinsic value ($260), a share of the company appears to be undervalued.

B is incorrect. The estimate of the intrinsic value is

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{\$5(1+0.04)}{(0.06-0.04)} = \$260 \text{ per share}$$

Given that the market value is lower than the estimate of the intrinsic value ($250 < $260), a share of the company appears to be undervalued, not overvalued. A share of a company is overvalued when the market value is higher than the intrinsic value.

A is incorrect. The estimate of the intrinsic value is

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{\$5(1+0.04)}{(0.06-0.04)} = \$260 \text{ per share}$$

Given that the market value is lower than the estimate of the intrinsic value ($250 < $260), a share of the company appears to be undervalued. A share of a company is fairly valued when the intrinsic value is equal to the market value. Using $5 as D_1, rather than D_0, would lead to the wrong conclusion that a share of the company is fairly valued:

$$\$5/(0.06 - 0.04) = \$250 \text{ per share}$$

Equity Valuation: Concepts and Basic Tools
LOS g
Section 4.2
Market Efficiency
LOS b
Section 2.2

99 What type of risk *most likely* affects an investor's ability to buy and sell bonds in the desired amounts and at the desired time?

A Market liquidity

B Spread

C Default

A is correct. The size of the spread between the bid price and the ask price is the primary measure of market liquidity of the issue. Market liquidity risk is the risk that the investor will have to sell a bond below its indicated value. The wider the bid–ask spread, the greater the market liquidity risk.

B is incorrect because spread risk is the risk that spreads widen.

C is incorrect because default risk is the risk the borrower defaults.

Fundamentals of Credit Analysis
LOS a
Section 2

100 Which of the following *most likely* exhibits negative convexity?

A A callable bond

B An option-free bond

C A putable bond

A is correct. A callable bond exhibits negative convexity at low yield levels and positive convexity at high yield levels.

B is incorrect because an option-free bond always exhibits positive convexity.

C is incorrect because a putable bond always exhibits positive convexity, higher than an option-free bond.

Understanding Fixed-Income Risk and Return
LOS h
Section 3.6

101 If a bank wants the ability to retire debt prior to maturity in order to take advantage of lower borrowing rates, it *most likely* issues a:

A convertible bond.

B callable bond.

C putable bond.

B is correct. Callable bonds give issuers the ability to retire debt prior to maturity. The most compelling reason for them to do so is to take advantage of lower borrowing rates.

A is incorrect because convertible bonds give bondholders the ability to convert the bond to a predetermined number of shares of the company.

C is incorrect because putable bonds give bondholders the ability to sell the bond at predetermined price to issuers.

Fixed-Income Markets: Issuance, Trading, and Funding
LOS g
Section 6.3.5

102 An investor purchases a 5% coupon bond maturing in 15 years for par value. Immediately after purchase, the yield required by the market increases. The investor would then *most likely* have to sell the bond at:

A a premium.

B a discount.

C par.

B is correct. The bond would sell below par or at a discount if the yield required by the market rises above the coupon rate. Because the bond initially was purchased at par, the coupon rate equals the yield required by the market. Subsequently, if yields rise above the coupon, the bond's market price would fall below par.

A is incorrect because the yield is now greater than the coupon.

C is incorrect because the yield is now greater than the coupon.

Introduction to Fixed-Income Valuation
LOS b
Section 2.3

103 The process of securitization is *least likely* to allow banks to:

A originate loans.

B reduce the layers between borrowers and ultimate investors.

C repackage loans into simpler structures.

C is correct. Securitization allows banks to originate (or create) loans, and the process results in a reduction in the layers between borrowers and ultimate investors. The loans are repackaged into more complex, not simpler, structures.

A is incorrect because securitization allows banks to originate (or create) loans.

B is incorrect because securitization results in a reduction in the layers between borrowers and ultimate investors.

Introduction to Asset-Backed Securities
LOS a
Section 2

104 Which of the following is *least likely* a component of yield spread?

A Taxation

B Expected inflation rate

C Credit risk

B is correct. Building blocks of the yield curve are spread (risk premium) and a benchmark (risk-free rate of return). Expected inflation rate and expected real rate are components of the risk-free rate of return (i.e., the benchmark).

A is incorrect because taxation is part of the yield spread providing the investor with compensation for the tax impact of holding a specific bond.

C is incorrect because credit risk is part of the yield spread providing the investor with compensation for the credit risks of holding a specific bond.

Introduction to Fixed-Income Valuation
LOS i
Section 5.1

105 A 90-day commercial paper issue is quoted at a discount rate of 4.75% for a 360-day year. The bond equivalent yield for this instrument is *closest* to:

A 4.87%.

B 4.81%.

C 4.75%.

A is correct. The price of the commercial paper per 100 of par value is

$$PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right)$$

where *PV* and *FV* are the price and face value of the money market instrument, Days is the number of days between settlement and maturity, Year is number of days in the year, and *DR* is the discount rate stated as an annual percentage.

$$PV = 100 \times \left(1 - \frac{90}{360} \times 0.0475\right) = 98.8125$$

The bond equivalent yield is

$$\begin{aligned} AOR &= \frac{\text{Year}}{\text{Days}} \times \left(\frac{FV - PV}{PV}\right) \\ &= \frac{365}{90} \times \left(\frac{100 - 98.8125}{98.8125}\right) \\ &= 4.874\% \end{aligned}$$

B is incorrect because the price of the commercial paper is incorrectly computed as

$$PV = 100 \times \left(1 - \frac{90}{365} \times 0.0475\right) = 98.8288$$

The bond equivalent yield is computed as

$$\begin{aligned} AOR &= \frac{\text{Year}}{\text{Days}} \times \left(\frac{FV - PV}{PV}\right) \\ &= \frac{365}{90} \times \left(\frac{100 - 98.8288}{98.8288}\right) \\ &= 4.806\% \end{aligned}$$

C is incorrect because it is just the discount rate itself.

Introduction to Fixed-Income Valuation
LOS f
Section 3.5

106 The factor *least likely* to influence the yield spread on an option-free, fixed-rate bond is a change in the:

A credit risk of the issuer.

B expected inflation rate.

C liquidity of the bond.

B is correct. For an option-free, fixed-rate bond, changes in the yield spread can arise from changes in the credit risk of the issuer and/or changes in the liquidity of the issue. Changes in the expected inflation rate influence the benchmark rate.

C is incorrect because changes in the yield spread an option-free, fixed-rate bond arise from changes in the liquidity of the issue.

A is incorrect because changes in the yield spread an option-free, fixed-rate bond arise from changes in the credit risk of the issuer.

Understanding Fixed-Income Risk and Return
LOS l
Section 5

107 Samsung Electronics Co has issued a five-year bond with a par value of $1,000 and a coupon rate of 6.5%. This bond is *most likely* to be classified as a:

A capital market security.

B surety bond.

C consol.

A is correct. Fixed-income securities with original maturities that are longer than one year are called capital market securities. The bond mentioned in the question has a five-year maturity and therefore is a capital market security.

B is incorrect because a surety bond is a bond that reimburses investors for any losses incurred if the issuer defaults.

C is incorrect because a consol has no stated maturity date.

Fixed-Income Securities: Defining Elements
LOS a
Section 2.1.2

108 In a mortgage pass-through security, the pass-through rate:

A is adjusted as market rates rise or fall.

B adjusts the rate on the underlying pool of mortgages by a servicing fee.

C is equal to the mortgage rate on the underlying pool of mortgages.

B is correct. In a mortgage pass-through security, the pass-through rate is less than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing (and other administrative) fees.

A is incorrect because in a mortgage pass-through security, the pass-through rate is less than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing (and other administrative) fees.

C is incorrect because in a mortgage pass-through security, the pass-through rate is less than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing (and other administrative) fees.

Introduction to Asset-Backed Securities
LOS e
Section 5.1.1

109 Which of the following is *most likely* an example of a Eurobond?

A A Canadian borrower issuing British pound–denominated bonds in the UK market.

B A Japanese borrower issuing US dollar–denominated bonds in the US market.

C An Australian borrower issuing Canadian dollar–denominated bonds in the UK market.

C is correct. A Eurobond is an international bond issued outside the jurisdiction of any one country and not denominated in the currency of the country where it is issued.

A is incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

B is incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

Fixed-Income Securities: Defining Elements
LOS d
Section 3.2

110 A bond is currently selling for 102.31. A valuation model estimates the price will fall to 101.12 if interest rates increase by 20 bps and rise to 103.74 if interest rates decrease by 20 bps. Using these estimates, the effective duration of the bond is *closest* to:

A 6.48.

B 6.40.

C 6.31.

B is correct. The effective duration of a bond is

$$\text{EffDur} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta\text{Curve}) \times (PV_0)}$$

where PV_-, PV_0, and PV_+ are the values of the bond when the yield falls, under the current yield, and when the yield rises, respectively, and ΔCurve is the change in the benchmark yield curve.

$$\text{EffDur} = \frac{103.74 - 101.12}{2 \times 102.31 \times 0.002} = 6.40$$

A is incorrect because it has 101.12 in the denominator instead of 102.31:

$$\frac{103.74 - 101.12}{2 \times 101.12 \times 0.002} = 6.48$$

C is incorrect because it has 103.74 in the denominator instead of 102.31:

$$\frac{103.74 - 101.12}{2 \times 103.74 \times 0.002} = 6.31$$

Understanding Fixed-Income Risk and Return
LOS b
Section 3.2

111 Which of the following derivatives is *least likely* to be classified as a contingent claim?

A A futures contract

B A call option contract

C A credit default swap

A is correct. A futures contract is classified as a forward commitment in which the buyer undertakes to purchase the underlying asset from the seller at a later date and at a price agreed on by the two parties when the contract is initiated.

B is incorrect. A call option contract is a contingent claim in which the buyer of the option has a right to purchase the underlying asset at a fixed price on or before a pre-specified expiration date.

C is incorrect. A credit default swap is a contingent claim in which the credit protection seller provides protection to the credit protection buyer against the credit risk of a third party.

Derivative Markets and Instruments
LOS b
Section 4.1.2

112 Forward rate agreements are *most likely* used to hedge an exposure in the:

A foreign exchange market.

B money market.

C equity market.

B is correct. Forward rate agreements are used to hedge interest rate exposure present in the money market.

A is incorrect. Forward rate agreements are used to hedge interest rate exposure and not foreign exchange exposure.

C is incorrect. Forward rate agreements are used to hedge interest rate exposure and not equity exposure.

Basics of Derivative Pricing and Valuation
LOS e
Section 3.1.4

113 An investor notices that the price of an American call option is above the price of a European call option with otherwise identical features. What is the *most likely* reason for this difference?

A The options are close to expiration.

B The options are deep in the money.

C The underlying will go ex-dividend.

C is correct. American call prices can differ from European call prices only if there are cash flows on the underlying.

A is incorrect. Early expiration of the option is not a reason for pricing differences between American and European call options. American call prices can differ from European call prices only if there are cash flows on the underlying.

B is incorrect. The fact that the option is deep in the money is not a reason for pricing differences between American and European call options. American call prices can differ from European call prices only if there are cash flows on the underlying.

Basics of Derivative Pricing and Valuation
LOS o
Section 4.3

114 If the exercise price of a European put option at expiration is below the price of the underlying, the value of the option is *most likely*:

A equal to zero.

B less than zero.

C greater than zero.

A is correct. If the exercise price of a European put option is below the underlying price at expiration, the option is worthless and has a value of zero.

B is incorrect. The value of an option can never be negative.

C is incorrect. For a positive value, exercise price must be below the price of underlying.

Basics of Derivative Pricing and Valuation
LOS i
Section 4.1.1

115 In efficient financial markets, risk-free arbitrage opportunities:

A will not exist.

B may persist in the long run.

C may exist temporarily.

C is correct. In efficient financial markets, risk-free arbitrage opportunities may exist temporarily, but their continuous exploitation will eliminate these arbitrage opportunities in the long run.

A is incorrect. Financial markets being efficient does not mean that risk-free arbitrage opportunities cannot exist.

B is incorrect. In efficient financial markets, any risk-free arbitrage opportunities will exist only temporarily because their continuous exploitation will result in these arbitrage opportunities being eliminated in the long run.

Derivative Markets and Instruments
LOS e
Section 7.2

116 Do management fees *most likely* get paid to the manager of a hedge fund, regardless of the fund's performance?

A No, only when the fund's net asset value exceeds the previous high-water mark

B No, only when the fund's gross return is positive

C Yes

C is correct. Regardless of performance, the management fee is always paid to the fund manager.

B is incorrect because the gross return can be at any level and the manager is still paid the management fee.

A is incorrect because the management fee is paid regardless of the value of the assets in the fund.

Introduction to Alternative Investments
LOS f
Section 3.3.1

117 Investors in alternative assets who seek liquidity are *most likely* to invest in:

A hedge funds.

B real estate investment trusts.

C private equity.

B is correct. Real estate investment trusts are publicly traded and thus provide liquidity.

A is incorrect. Hedge funds may have long lockup periods.

C is incorrect. Private equity funds may have long lockup periods.

Introduction to Alternative Investments
LOS d
Section 8.1.1

118 The real estate valuation method that uses a discounted cash flow model is *best* characterized as:

A a comparable sales approach.

B a cost approach.

C an income approach.

C is correct. The income approach to real estate valuation values a property by using a discounted cash flow model.

A is incorrect. The comparable sales approach involves determining a value based on recent sales of similar properties.

B is incorrect. The cost approach evaluates the replacement cost of the property.

Introduction to Alternative Investments
LOS e
Section 5.4

119 Which of the following is *least likely* to reduce the likelihood of being defrauded by a dishonest money manager?

A Third-party custody of assets under management

B Strong and consistent reported investment performance

C Independent verification of investment results

B is correct. To prevent fraud, involvement of third parties in the reporting and asset management process is helpful. A strong and consistent reported investment performance that lacks outside verification may actually be a warning sign.

A is incorrect. Third-party custody of assets under management helps to reduce the possibility of fraud.

C is incorrect. Independent verification of investment results helps to reduce the possibility of fraud.

Introduction to Alternative Investments
LOS g
Section 8.3

120 In valuing underlying hedge fund positions, the most conservative approach is *most likely* one that uses:

A the average of the bid and ask prices.

B bid prices for longs and ask prices for shorts.

C the most recent market prices.

B is correct A conservative and theoretically accurate approach is to use bid prices for longs and ask prices for shorts because these are the prices at which the positions could be closed.

A is incorrect because although using the average quote [(bid + ask)/2] is a common approach, a more conservative and theoretically accurate approach is to use bid prices for longs and ask prices for shorts as these are the prices at which the positions could be closed.

C is incorrect because when market prices or quotes are used for valuation, funds may differ in which price or quote they use (for example, bid price, ask price, average quote, and median quote).

Introduction to Alternative Investments
LOS d
Section 3.4